

CHAPTER C13

SEISMIC DESIGN REQUIREMENTS FOR NONSTRUCTURAL COMPONENTS

C13.1 GENERAL

Chapter 13 defines minimum design criteria for architectural, mechanical, electrical, and other nonstructural systems and components, recognizing structure use, occupant load, the need for operational continuity, and the interrelation of structural, architectural, mechanical, electrical, and other nonstructural components. Nonstructural components are designed for design earthquake ground motions as defined in Section 11.2 and determined in Section 11.4.5 of the standard. In contrast to structures, which are implicitly designed for a low probability of collapse when subjected to the risk-targeted maximum considered earthquake (MCE_R) ground motions, there are no implicit performance goals associated with the MCE_R for nonstructural components. Performance goals associated with the design earthquake are discussed in Section C13.1.3.

Suspended or attached nonstructural components that could detach either in full or in part from the structure during an earthquake are referred to as falling hazards and may represent a serious threat to property and life safety. Critical attributes that influence the hazards posed by these components include their weight, their attachment to the structure, their failure or breakage characteristics (e.g., nonshatterproof glass), and their location relative to occupied areas (e.g., over an entry or exit, a public walkway, an atrium, or a lower adjacent structure). Architectural components that pose potential falling hazards include parapets, cornices, canopies, marquees, glass, large ornamental elements (e.g., chandeliers), and building cladding. In addition, suspended mechanical and electrical components (e.g., mixing boxes, piping, and ductwork) may represent serious falling hazards. Figs. C13.1-1 through C13.1-4 show damage to nonstructural components in past earthquakes.

Components whose collapse during an earthquake could result in blockage of the means of egress deserve special consideration. The term “means of egress” is used commonly in building codes with respect to fire hazard. Egress paths may include intervening aisles, doors, doorways, gates, corridors, exterior exit balconies, ramps, stairways, pressurized enclosures, horizontal exits, exit passageways, exit courts, and yards. Items whose failure could jeopardize the means of egress include walls around stairs and corridors, veneers, cornices, canopies, heavy partition systems, ceilings, architectural soffits, light fixtures, and other ornaments above building exits or near fire escapes. Examples of components that generally do not pose a significant falling hazard include fabric awnings and canopies. Architectural, mechanical, and electrical components that, if separated from the structure, fall in areas that are not accessible to the public (e.g., into a mechanical shaft or light well) also pose little risk to egress routes.

For some architectural components, such as exterior cladding elements, wind design forces may exceed the calculated seismic

design forces. Nevertheless, seismic detailing requirements may still govern the overall structural design. Where this is a possibility, it must be investigated early in the structural design process.

The seismic design of nonstructural components may involve consideration of nonseismic requirements that are affected by seismic bracing. For example, accommodation of thermal expansion in pressure piping systems often is a critical design consideration, and seismic bracing for these systems must be arranged in a manner that accommodates thermal movements. Particularly in the case of mechanical and electrical systems, the design for seismic loads should not compromise the functionality, durability, or safety of the overall design; this method requires collaboration among the various disciplines of the design and construction team.

For various reasons (e.g., business continuity), it may be desirable to consider higher performance than that required by the building code. For example, to achieve continued operability of a piping system, it is necessary to prevent unintended operation of valves or other in-line components in addition to preventing collapse and providing leak tightness. Higher performance also is required for components containing substantial quantities of hazardous contents (as defined in Section 11.2). These components must be designed to prevent uncontrolled release of those materials.

The requirements of Chapter 13 are intended to apply to new construction and tenant improvements installed at any time during the life of the structure, provided that they are listed in Table 13.5-1 or 13.6-1. Furthermore, they are intended to reduce (not eliminate) the risk to occupants and to improve the likelihood that essential facilities remain functional. Although property protection (in the sense of investment preservation) is a possible consequence of implementation of the standard, it is not currently a stated or implied goal; a higher level of protection may be advisable if such protection is desired or required.

C13.1.1 Scope. The requirements for seismic design of nonstructural components apply to the nonstructural component and to its supports and attachments to the main structure. In some cases, as defined in Section 13.2, it is necessary to consider explicitly the performance characteristics of the component. The requirements are intended to apply only to permanently attached components, not to furniture, temporary items, or mobile units. Furniture, such as tables, chairs, and desks, may shift during strong ground shaking but generally poses minimal hazards provided that it does not obstruct emergency egress routes. Storage cabinets, tall bookshelves, and other items of significant mass do not fall into this category and should be anchored or braced in accordance with this chapter.

Temporary items are those that remain in place for short periods of time (months, not years). Components that are



FIGURE C13.1-1. Hospital Imaging Equipment That Fell from Overhead Mounts



FIGURE C13.1-2 Damaged Ceiling System



FIGURE C13.1-3 Collapsed Light Fixtures



FIGURE C13.1-4 Toppled Storage Cabinets



FIGURE C13.1-5 Collapsed Duct and HVAC Diffuser



FIGURE C13.1-6 Skid-Mounted Components

expected to remain in place for periods of a year or longer, even if they are designed to be movable, should be considered permanent for the purposes of this section. Modular office systems are considered permanent because they ordinarily remain in place for long periods. In addition, they often include storage units that have significant capacity and may topple in an earthquake. They are subject to the provisions of Section 13.5.8 for partitions if

they exceed 6 ft (1.8 m) high. Mobile units include components that are moved from one point in the structure to another during ordinary use. Examples include desktop computers, office equipment, and other components that are not permanently attached to the building utility systems (Fig. C13.1-5). Components that are mounted on wheels to facilitate periodic maintenance or cleaning

but that otherwise remain in the same location (e.g., server racks) are not considered movable for the purposes of anchorage and bracing. Likewise, skid-mounted components (as shown in Fig. C13.1-6), as well as the skids themselves, are considered permanent equipment.

With the exception of solar panels satisfying the provisions of Section 13.6.12, equipment must be anchored if it is permanently attached to utility services (electricity, gas, and water). For the purposes of this requirement, “permanently attached” should be understood to include all electrical connections except NEMA 5-15 and 5-20 straight-blade connectors (duplex receptacles).

C13.1.2 Seismic Design Category. The requirements for nonstructural components are based in part on the Seismic Design Category (SDC) to which they are assigned. As the SDC is established considering factors not unique to specific nonstructural components, all nonstructural components occupying or attached to a structure are assigned to the same SDC as the structure.

C13.1.3 Component Importance Factor. Performance expectations for nonstructural components often are defined in terms of the functional requirements of the structure to which the components are attached. Although specific performance goals for nonstructural components have yet to be defined in building codes, the component Importance Factor (I_p) implies performance levels for specific cases. For noncritical nonstructural components (those with a component Importance Factor, I_p , of 1.0), the following behaviors are anticipated for shaking of different levels of intensity:

1. Minor earthquake ground motions—minimal damage; not likely to affect functionality;
2. Moderate earthquake ground motions—some damage that may affect functionality; and
3. Design earthquake ground motions—major damage but significant falling hazards are avoided; likely loss of functionality.

Components with Importance Factors greater than 1.0 are expected to remain in place, sustain limited damage, and when necessary, function after an earthquake (see Section C13.2.2). These components can be located in structures that are not assigned to Risk Category IV. For example, fire sprinkler piping systems have an Importance Factor, I_p , of 1.5 in all structures because these essential systems should function after an earthquake. Egress stairways are assigned an I_p of 1.5 as well, although in many cases the design of these stairways is dictated by differential displacements, not inertial force demands.

The component Importance Factor is intended to represent the greater of the life-safety importance of the component and the hazard-exposure importance of the structure. It indirectly influences the survivability of the component via required design forces and displacement levels, as well as component attachments and detailing. Although this approach provides some degree of confidence in the seismic performance of a component, it may not be sufficient in all cases. For example, individual ceiling tiles may fall from a ceiling grid that has been designed for larger forces. This problem may not represent a serious falling hazard if the ceiling tiles are made of lightweight materials, but it may lead to blockage of critical egress paths or disruption of the facility function. When higher levels of confidence in performance are required, the component is classified as a designated seismic system (Section 11.2), and in certain cases, seismic qualification of the component or system is necessary. Seismic qualification approaches are provided in Sections 13.2.5 and 13.2.6. In addition, seismic qualification approaches presently in use by the Department of Energy (DOE) can be applied.

Risk Category IV structures are intended to be functional after a design earthquake; critical nonstructural components and

equipment in such structures are designed with I_p equal to 1.5. This requirement applies to most components and equipment because damage to vulnerable unbraced systems or equipment may disrupt operations after an earthquake even if they are not directly classified as essential to life safety. The nonessential and nonhazardous components are themselves not affected by this requirement. Instead, requirements focus on the supports and attachments. UFC 3-310-04 (DOD 2007) has additional guidance for improved performance.

C13.1.4 Exemptions. Several classes of nonstructural components are exempted from the requirements of Chapter 13. The exemptions are made on the assumption that, either because of their inherent strength and stability or the lower level of earthquake demand (accelerations and relative displacements), or both, these nonstructural components and systems can achieve the performance goals described earlier in this commentary without explicitly satisfying the requirements of this chapter.

The requirements are intended to apply only to permanent components, not furniture and temporary or mobile equipment. Furniture (with the exception of more massive elements like storage cabinets) may shift during strong ground shaking but poses minimal hazards. With the exception of solar panels satisfying the provisions of Section 13.6.12, equipment must be anchored if it is permanently attached to the structure utility services, such as electricity, gas, or water. For the purposes of this requirement, “permanently attached” includes all electrical connections except plugs for duplex receptacles.

Temporary items are those that remain in place for six months or less. Modular office systems are considered permanent since they ordinarily remain in place for long periods. In addition, they often include storage units of significant capacity, which may topple in earthquakes. Mobile units include components that are moved from one point in the structure to another during ordinary use. Examples include desktop computers, office equipment, and other components that are not permanently attached to the building utility systems. Components mounted on wheels to facilitate periodic maintenance or cleaning but that otherwise remain in the same location are not considered movable for the purposes of anchorage and bracing.

Furniture resting on floors, such as tables, chairs, and desks, may shift during strong ground shaking, but they generally pose minimal hazards, provided that they do not obstruct emergency egress routes. Examples also include desktop computers, office equipment, and other components that are not permanently attached to the building utility systems.

With the exception of parapets supported by bearing walls or shear walls, all components in Seismic Design Categories A and B are exempt because of the low levels of ground shaking expected. Parapets are not exempt because experience has shown that these items can fail and pose a significant falling hazard, even at low-level shaking levels.

Discrete components are generally understood to be stand-alone items such as cabinets, pumps, electrical boxes, lighting, and signage. Discrete components, architectural or mechanical, weighing 20 lb (89 N) or less generally do not pose a risk and are exempted provided that they are positively attached to the structure, regardless of whether they carry an Importance Factor, I_p , of 1.5 or not. Larger items up to 400 lb (1,780 N) in weight with $I_p = 1.0$ have historically been exempted provided that they are positively attached and have flexible connections. The exemption for mechanical and electrical components in Seismic Design Categories D, E, or F based on weight and location of the center of mass is particularly applicable to vertical equipment racks and similar components. Where detailed information regarding the center of mass of the intended installation is unavailable, a conservative estimate based on potential equipment configurations



FIGURE C13.1-7 Premanufactured Modular Mechanical Systems

Source: Courtesy of Matthew Tobolski.

should be used. The exemption for components weighing 400 lb (1,780 N) or less has existed in provisions for nonstructural components for many years and corresponds roughly to the weight of a 40-gal. (150-L) hot water tank. Coupled with this and the other exemptions in SDC D, E, and F is a requirement that the component be positively attached to the structure. Positive attachment is provided when the attachment is carried out using appropriate structural-grade materials whereby explicit design calculations for the anchorage are not required.

Although the exemptions listed in Section 13.1.4 are intended to waive bracing requirements for nonstructural components that are judged to pose negligible life-safety hazard, in some cases it may nevertheless be advisable to consider bracing (in consultation with the owner) for exempted components to minimize repair costs and/or disproportionate loss (e.g., art works of high value).

The bracing exemptions for short hangers have been moved to the respective sections in which they apply. These exemptions are based on the assumption that the hangers have sufficient ductility to undergo plastic deformations without failure while at the same time providing sufficient stiffness to limit lateral displacement to a reasonable level. This assumption extends to the anchors, and as such the design and detailing of the connections to the structure should take this into account. Raceways, ducts, and piping systems must be able to accommodate the relative displacement demands calculated in Section 13.3.2, since these displacements can be substantially greater than those that occur at connections to equipment. At seismic separation joints between structures, large displacements may occur over a short distance.

Short hangers fabricated from threaded rods resist lateral force primarily through bending and are prone to failure through cyclic fatigue. Tests conducted by Soulages and Weir (2011) suggest that low cycle fatigue is not an issue when the ductility ratios for the rods are less than about 4. The testing also indicated that swivel connections are not required, provided that the load and rod length limitations are observed. The limits on unbraced trapezes and hangers are based on limiting the ductility ratios to reasonable levels, when subject to the maximum force demands in the highest seismic risk regions. It should be noted that in areas of lower seismic risk, less restrictive criteria could be used.

The exemption for short hangers is limited to the case where every hanger in the raceway run is less than 12 in. (305 mm) because of the need to carefully consider the seismic loads and compatible displacement limits for the portions of raceways with longer hanger supports.

The historical exemption for trapeze-supported conduit less than 2.5 in. (64 mm) trade size has been removed, since its application to specific cases, such as a trapeze supporting multiple conduit runs, was unclear.

The exemption for trapezes with short rod hangers applies only to trapezes configured with the rod hangers attached directly to the trapeze and the structural framing. Where one or more rod hangers for a trapeze are supported from another trapeze, the bracing exemption does not apply.

C13.1.5 Premanufactured Modular Mechanical and Electrical Systems. Large premanufactured modular mechanical and electrical systems (as shown in Fig. C13.1-7) should be considered nonbuilding structures for the purposes of the enveloping structural system design, unless the module has been prequalified in accordance with Section 13.2.2. However, where the premanufactured module has not been prequalified, the nonstructural components contained within the module should be addressed through the requirements of Chapter 13. Note that this provision is not intended to address skid-mounted equipment assemblies not equipped with an enclosure, nor does it address single large components, such as air handlers, cooling towers, chillers, and boilers.

C13.1.6 Application of Nonstructural Component Requirements to Nonbuilding Structures. At times, a nonstructural component should be treated as a nonbuilding structure. When the physical characteristics associated with a given class of nonstructural components vary widely, judgment is needed to select the appropriate design procedure and coefficients. For example, cooling towers vary from small packaged units with an operating weight of 2,000 lb (8.9 kN) or less to structures the size of buildings. Consequently, design coefficients for the design of “cooling towers” are found both in Tables 13.6-1 and 15.4-2. Small cooling towers are best designed as nonstructural components using the provisions of Chapter 13, whereas large ones are clearly nonbuilding structures that are more appropriately designed using the provisions of Chapter 15. Similar issues arise for other classes of nonstructural component (e.g., boilers and bins). Guidance on determining whether an item should be treated as a nonbuilding structure or nonstructural component for the purpose of seismic design is provided in Bachman and Dowty (2008).

The specified weight limit for nonstructural components (25% relative to the combined weight of the structure and component) relates to the condition at which dynamic interaction between the component and the supporting structural system is potentially significant. Section 15.3.2 contains requirements for addressing this interaction in design.

C13.1.7 Reference Documents. Professional and trade organizations have developed nationally recognized codes and standards for the design and construction of specific mechanical

and electrical components. These documents provide design guidance for normal and upset (abnormal) operating conditions and for various environmental conditions. Some of these documents include earthquake design requirements in the context of the overall mechanical or electrical design. It is the intent of the standard that seismic requirements in referenced documents be used. The developers of these documents are familiar with the expected performance and failure modes of the components; however, the documents may be based on design considerations not immediately obvious to a structural design professional. For example, in the design of industrial piping, stresses caused by seismic inertia forces typically are not added to those caused by thermal expansion.

Where reference documents have been adopted specifically by this standard as meeting the force and displacement requirements of this chapter with or without modification, they are considered to be a part of the standard.

There is a potential for misunderstanding and misapplication of reference documents for the design of mechanical and electrical systems. A registered design professional familiar with both the standard and the reference documents used should be involved in the review and acceptance of the seismic design.

Even when reference documents for nonstructural components lack specific earthquake design requirements, mechanical and electrical equipment constructed in accordance with industry-standard reference documents have performed well historically when properly anchored. Nevertheless, manufacturers of mechanical and electrical equipment are expected to consider seismic loads in the design of the equipment itself, even when such consideration is not explicitly required by this chapter.

Although some reference documents provide requirements for seismic capacity appropriate to the component being designed, the seismic demands used in design may not be less than those specified in the standard.

Specific guidance for selected mechanical and electrical components and conditions is provided in Section 13.6.

Unless exempted in Section 13.1.4, components should be anchored to the structure and, to promote coordination, required supports and attachments should be detailed in the construction documents. Reference documents may contain explicit instruction for anchorage of nonstructural components. The anchorage requirements of Section 13.4 must be satisfied in all cases, however, to ensure a consistent level of robustness in the attachments to the structure.

C13.1.8 Reference Documents Using Allowable Stress Design. Many nonstructural components are designed using specifically developed reference documents that are based on allowable stress loads and load combinations and generally permit increases in allowable stresses for seismic loading. Although Section 2.4.1 of the standard does not permit increases in allowable stresses, Section 13.1.8 explicitly defines the conditions for stress increases in the design of nonstructural components where reference documents provide a basis for earthquake-resistant design.

C13.2 GENERAL DESIGN REQUIREMENTS

C13.2.1 Applicable Requirements for Architectural, Mechanical, and Electrical Components, Supports, and Attachments. Compliance with the requirements of Chapter 13 may be accomplished by project-specific design or by a manufacturer's certification of seismic qualification of a system or component. When compliance is by manufacturer's certification, the items must be installed in accordance with the manufacturer's

requirements. Evidence of compliance may be provided in the form of a signed statement from a representative of the manufacturer or from the registered design professional indicating that the component or system is seismically qualified. One or more of the following options for evidence of compliance may be applicable:

1. An analysis (e.g., of a distributed system such as piping) that includes derivation of the forces used for the design of the system, the derivation of displacements and reactions, and the design of the supports and anchorages;
2. A test report, including the testing configuration and boundary conditions used (where testing is intended to address a class of components, the range of items covered by the testing performed should also include the justification of similarities of the items that make this certification valid); and/or
3. An experience data report.

Components addressed by the standard include individual simple units and assemblies of simple units for which reference documents establish seismic analysis or qualification requirements. Also addressed by the standard are complex architectural, mechanical, and electrical systems for which reference documents either do not exist or exist for only elements of the system. In the design and analysis of both simple components and complex systems, the concepts of flexibility and ruggedness often can assist the designer in determining the necessity for analysis and, when analysis is necessary, the extent and methods by which seismic adequacy may be determined. These concepts are discussed in Section C13.6.1.

C13.2.2 Special Certification Requirements for Designed Seismic Systems. This section addresses the qualification of active designated seismic equipment, its supports, and attachments with the goals of improving survivability and achieving a high level of confidence that a facility will be functional after a design earthquake. Where components are interconnected, the qualification should provide the permissible forces (e.g., nozzle loads) and, as applicable, anticipated displacements of the component at the connection points to facilitate assessment for consequential damage, in accordance with Section 13.2.3. Active equipment has parts that rotate, move mechanically, or are energized during operation. Active designated seismic equipment constitutes a limited subset of designated seismic systems. Failure of active designated seismic equipment itself may pose a significant hazard. For active designated seismic equipment, failure of structural integrity and loss of function are to be avoided.

Examples of active designated seismic equipment include mechanical (components of HVACR systems and piping systems) or electrical (power supply distribution) equipment, medical equipment, fire pump equipment, and uninterruptible power supplies for hospitals. It is generally understood that fire protection sprinkler piping systems designed and installed per NFPA 13 are deemed to comply with the special certification requirements of Section 13.2.2. See Section 13.6.7.2.

There are practical limits on the size of a component that can be qualified via shake table testing. Components too large to be qualified by shake table testing need to be qualified by a combination of structural analysis and qualification testing or empirical evaluation through a subsystem approach. Subsystems of large, complex components (e.g., large chillers, skid-mounted equipment assemblies, and boilers) can be qualified individually, and the overall structural frame of the component can be evaluated by structural analysis.

Evaluating postearthquake operational performance for active equipment by analysis generally involves sophisticated modeling

with experimental validation and may not be reliable. Therefore, the use of analysis alone for active or energized components is not permitted unless a comparison can be made to components that have been otherwise deemed as rugged. As an example, a transformer is energized but contains components that can be shown to remain linearly elastic and are inherently rugged. However, switch equipment that contains fragile components is similarly energized but not inherently rugged, and it therefore cannot be certified solely by analysis. For complex components, testing or experience may therefore be the only practical way to ensure that the equipment will be operable after a design earthquake. Past earthquake experience has shown that much active equipment is inherently rugged. Therefore, evaluation of experience data, together with analysis of anchorage, is adequate to demonstrate compliance of active equipment such as pumps, compressors, and electric motors. In other cases, such as for motor control centers and switching equipment, shake table testing may be required.

With some exceptions (e.g., elevator motors), experience indicates that active mechanical and electrical components that contains electric motors of greater than 10 hp (7.4 kW) or that have a thermal exchange capacity greater than 200 MBH are unlikely to merit the exemption from shake table testing on the basis of inherent ruggedness. Components with lesser motor horsepower and thermal exchange capacity are generally considered to be small active components and are deemed rugged. Exceptions to this rule may be appropriate for specific cases, such as elevator motors that have higher horsepower but have been shown by experience to be rugged. Analysis is still required to ensure the structural integrity of the nonactive components. For example, a 15-ton condenser would require analysis of the load path between the condenser fan and the coil to the building structure attachment.

Where certification is accomplished by analysis, the type and sophistication of the required analysis varies by specific equipment type and construction. Static analysis using the total force specified in Section 13.3 considering applicable load combinations may be appropriate for single components where the structural frame is the only item to be certified and where internal dynamic effects are shown to be negligible. For single components where dynamic effects may be significant, or for assemblies of components, dynamic analysis is strongly suggested. Either modal analysis or response history procedures may be used, but care should be exercised when using modal analysis to ensure that the significant interactions between individual components are properly captured. In all analyses, it is essential that the stiffness, mass, and applied load be distributed in accordance with the component properties, and in sufficient detail (number of degrees of freedom) to allow for the desired forces, deformations, and accelerations to be accurately determined. Input motions for dynamic procedures should reflect the expected motion at the attachment points of the component. Nonlinear behavior of the component is typically not advisable in the certification analysis in the absence of well-documented test results for specific components. Generally, the input motion is (a) a generic floor response spectrum such as that provided in the ICC-ES AC 156, (b) location- and structure-specific floor spectra generated using the procedures of Section 13.3.1, or (c) acceleration time histories developed using dynamic analysis procedures similar to those specified in Chapter 16 or Section 12.9. Horizontal and vertical inputs are usually applied simultaneously when performing these types of dynamic analyses. As with all structural analysis, judgment is required to ensure that the results are applicable and representative of the behavior anticipated for the input motions.

C13.2.3 Consequential Damage. Although the components identified in Tables 13.5-1 and 13.6-1 are listed separately, significant interrelationships exist and must be considered. Consequential damage occurs because of interaction between components and systems. Even “braced” components displace, and the displacement between lateral supports can be significant in the case of distributed systems such as piping systems, cable and conduit systems, and other linear systems. It is the intent of the standard that the seismic displacements considered include both relative displacement between multiple points of support (addressed in Section 13.3.2) and, for mechanical and electrical components, displacement within the component assemblies. Impact of components must be avoided, unless the components are fabricated of ductile materials that have been shown to be capable of accommodating the expected impact loads. With protective coverings, ductile mechanical and electrical components and many more fragile components are expected to survive all but the most severe impact loads. Flexibility and ductility of the connections between distribution systems and the equipment to which they attach is essential to the seismic performance of the system.

The determination of the displacements that generate these interactions is not addressed explicitly in Section 13.3.2.1. That section concerns relative displacement of support points. Consequential damage may occur because of displacement of components and systems between support points. For example, in older suspended ceiling installations, excessive lateral displacement of a ceiling system may fracture sprinkler heads that project through the ceiling. A similar situation may arise if sprinkler heads projecting from a small-diameter branch line pass through a rigid ceiling system. Although the branch line may be properly restrained, it may still displace sufficiently between lateral support points to affect other components or systems. Similar interactions occur where a relatively flexible distributed system connects to a braced or rigid component.

The potential for impact between components that are in contact with or close to other structural or nonstructural components must be considered. However, where considering these potential interactions, the designer must determine if the potential interaction is both credible and significant. For example, the fall of a ceiling panel located above a motor control center is a credible interaction because the falling panel in older suspended ceiling installations can reach and impact the motor control center. An interaction is significant if it can result in damage to the target. Impact of a ceiling panel on a motor control center may not be significant because of the light weight of the ceiling panel. Special design consideration is appropriate where the failure of a nonstructural element could adversely influence the performance of an adjacent critical nonstructural component, such as an emergency generator.

C13.2.4 Flexibility. In many cases, flexibility is more important than strength in the performance of distributed systems, such as piping and ductwork. A good understanding of the displacement demand on the system, as well as its displacement capacity, is required. Components or their supports and attachments must be flexible enough to accommodate the full range of expected differential movements; some localized inelasticity is permitted in accommodating the movements. Relative movements in all directions must be considered. For example, even a braced branch line of a piping system may displace, so it needs to be connected to other braced or rigid components in a manner that accommodates the displacements without failure (Fig. C13.2-1). A further example is provided by cladding units (such as precast concrete

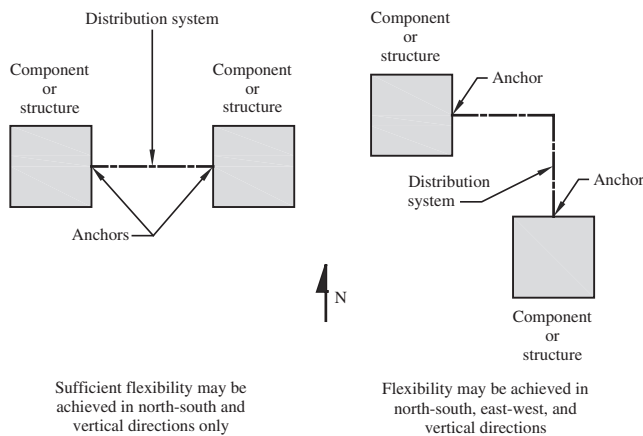


FIGURE C13.2-1 Schematic Plans Illustrating Branch Line Flexibility

wall units). Often very rigid in plane, cladding units require connections capable of accommodating story drift if attached at more than one level. (See Fig. C13.3-4 for an illustration.)

If component analysis assumes rigid anchors or supports, the predicted loads and local stresses can be unrealistically large, so it may be necessary to consider anchor and/or support stiffness.

C13.2.5 Testing Alternative for Seismic Capacity Determination. Testing is a well-established alternative method of seismic qualification for small- to medium-size equipment. Several national reference documents have testing requirements adaptable for seismic qualification. One such reference document (ICC-ES AC 156) is a shake table testing protocol that has been adopted by the International Code Council Evaluation Service. It was developed specifically to be consistent with acceleration demands (that is, force requirements) of the standard.

The development or selection of testing and qualification protocols should at a minimum include the following:

1. Description of how the protocol meets the intent for the project-specific requirements and relevant interpretations of the standard;
2. Definition of a test input motion with a response spectrum that meets or exceeds the design earthquake spectrum for the site;
3. Accounting for dynamic amplification caused by above-grade equipment installations (consideration of the actual dynamic characteristics of the primary support structure is permitted, but not required);
4. Definition of how shake table input demands were derived;
5. Definition and establishment of a verifiable pass/fail acceptance criterion for the seismic qualification based on the equipment Importance Factor and consistent with the building code and project-specific design intent; and
6. Development of criteria that can be used to rationalize test unit configuration requirements for highly variable equipment product lines.

To aid the design professional in assessing the adequacy of the manufacturer's certificate of compliance, it is recommended that certificates of compliance include the following:

1. Product family or group covered;
2. Building code(s) and standard(s) for which compliance was evaluated;
3. Testing standard used;

4. Performance objective and corresponding Importance Factor ($I_p = 1.0$ or $I_p = 1.5$);
5. Seismic demand for which the component is certified, including code and/or standard design parameters used to calculate seismic demand (such as values used for a_p , R_p , and site class); and
6. Installation restrictions, if any (grade, floor, or roof level).

Without a test protocol recognized by the building code, qualification testing is inconsistent and difficult to verify. The use of ICC-ES AC 156 simplifies the task of compliance verification because it was developed to address directly the testing alternative for nonstructural components, as specified in the standard. It also sets forth minimum test plan and report deliverables.

Use of other standards or ad hoc protocols to verify compliance of nonstructural components with the requirement of the standard should be considered carefully and used only where project-specific requirements cannot be met otherwise.

Where other qualification test standards are used, in whole or in part, it is necessary to verify compliance with this standard. For example, IEEE 693 (2005) indicates that it is to be used for the sole purpose of qualifying electrical equipment (specifically listed in the standard) for use in utility substations. Where equipment testing has been conducted to other standards (for instance, testing done in compliance with IEEE 693), a straightforward approach would be to permit evaluation, by the manufacturer, of the test plan and data to validate compliance with the requirements of ICC-ES AC 156 because it was developed specifically to comply with the seismic demands of this standard.

The qualification of mechanical and electrical components for seismic loads alone may not be sufficient to achieve high-performance objectives. Establishing a high confidence that performance goals will be met requires consideration of the performance of structures, systems (e.g., fluid, mechanical, electrical, and instrumentation), and their interactions (e.g., interaction of seismic and other loads), as well as compliance with installation requirements.

C13.2.6 Experience Data Alternative for Seismic Capacity Determination. An established method of seismic qualification for certain types of nonstructural components is the assessment of data for the performance of similar components in past earthquakes. The seismic capacity of the component in question is extrapolated based on estimates of the demands (e.g., force or displacement) to which the components in the database were subjected. Procedures for such qualification have been developed for use in nuclear facility applications by the Seismic Qualification Utility Group (SQUG) of the Electric Power Research Institute.

The SQUG rules for implementing the use of experience data are described in a proprietary Generic Implementation Procedure database. It is a collection of findings from detailed engineering studies by experts for equipment from a variety of utility and industrial facilities.

Valid use of experience data requires satisfaction of rules that address physical characteristics; manufacturer's classification and standards; and findings from testing, analysis, and expert consensus opinion.

Four criteria are used to establish seismic qualification by experience, as follows:

1. Seismic capacity versus demand (a comparison with a bounding spectrum);
2. Earthquake experience database cautions and inclusion rules;

3. Evaluation of anchorage; and
4. Evaluation of seismic interaction.

Experience data should be used with care because the design and manufacture of components may have changed considerably in the intervening years. The use of this procedure is also limited by the relative rarity of strong-motion instrument records associated with corresponding equipment experience data.

C13.2.7 Construction Documents. Where the standard requires seismic design of components or their supports and attachments, appropriate construction documents defining the required construction and installation must be prepared. These documents facilitate the special inspection and testing needed to provide a reasonable level of quality assurance. Of particular concern are large nonstructural components (such as rooftop chillers) whose manufacture and installation involve multiple trades and suppliers and which impose significant loads on the supporting structure. In these cases, it is important that the construction documents used by the various trades and suppliers be prepared by a registered design professional to satisfy the seismic design requirements.

The information required to prepare construction documents for component installation includes the dimensions of the component, the locations of attachment points, the operating weight, and the location of the center of mass. For instance, if an anchorage angle is attached to the side of a metal chassis, the gauge and material of the chassis must be known so that the number and size of required fasteners can be determined. Or when a piece of equipment has a base plate that is anchored to a concrete slab with expansion anchors, the drawings must show the base plate's material and thickness, the diameter of the bolt holes in the plate, and the size and depth of embedment of the anchor bolts. If the plate is elevated above the slab for leveling, the construction documents must also show the maximum gap permitted between the plate and the slab.

C13.3 SEISMIC DEMANDS ON NONSTRUCTURAL COMPONENTS

The seismic demands on nonstructural components, as defined in this section, are acceleration demands and relative displacement demands. Acceleration demands are represented by equivalent static forces. Relative displacement demands are provided directly and are based on either the actual displacements computed for the structure or the maximum allowable drifts that are permitted for the structure.

C13.3.1 Seismic Design Force. The seismic design force for a component depends on the weight of the component, the component Importance Factor, the component response modification factor, the component amplification factor, and the component acceleration at a point of attachment to the structure. The forces prescribed in this section of the standard reflect the dynamic and structural characteristics of nonstructural components. As a result of these characteristics, forces used for verification of component integrity and design of connections to the supporting structure typically are larger than those used for design of the overall seismic force-resisting system.

Certain nonstructural components lack the desirable attributes of structures (such as ductility, toughness, and redundancy) that permit the use of greatly reduced lateral design forces. Thus values for the response modification factor, R_p , in Tables 13.5-1 and 13.6-1 generally are smaller than R values for structures. These R_p values, used to represent the energy absorption capability of a component and its attachments, depend on both

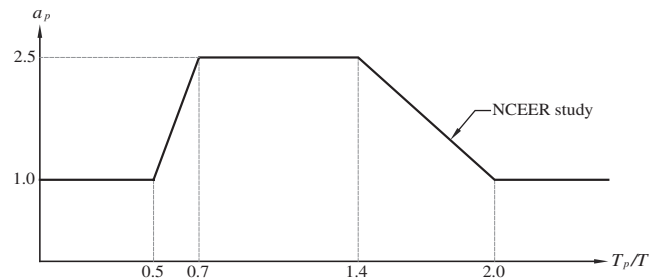


FIGURE C13.3-1 NCEER Formulation for a_p as Function of Structural and Component Periods

overstrength and deformability. At present, these potentially separate considerations are combined in a single factor. The tabulated values are based on the collective judgment of the responsible committee.

Beginning with the 2005 edition of ASCE 7, significant adjustments have been made to tabulated R_p values for certain mechanical and electrical systems. For example, the value of R_p for welded steel piping systems is increased from 3.5 to 9. The a_p value increased from 1.0 to 2.5, so although it might appear that forces on such piping systems have been reduced greatly, the net change is negligible because R_p/a_p changes from 3.5 to 3.6. The minimum seismic design force of Eq. (13.3-3), which governs in many cases, is unchanged.

The component amplification factor (a_p) represents the dynamic amplification of component responses as a function of the fundamental periods of the structure (T) and component (T_p). When components are designed or selected, the structural fundamental period is not always defined or readily available. The component fundamental period (T_p) is usually only accurately obtained by shake table or pull-back tests and is not available for the majority of components. Tabulated a_p values are based on component behavior that is assumed to be either rigid or flexible. Where the fundamental period of the component is less than 0.06 s, dynamic amplification is not expected and the component is considered rigid. The tabulation of assumed a_p values is not meant to preclude more precise determination of the component amplification factor where the fundamental periods of both structure and component are available. The National Center for Earthquake Engineering Research formulation shown in Fig. C13.3-1 may be used to compute a_p as a function of T_p/T .

Dynamic amplification occurs where the period of a nonstructural component closely matches that of any mode of the supporting structure, although this effect may not be significant depending on the ground motion. For most buildings, the primary mode of vibration in each direction has the most influence on the dynamic amplification for nonstructural components. For long-period structures (such as tall buildings), where the period of vibration of the fundamental mode is greater than 3.5 times T_s , higher modes of vibration may have periods that more closely match the period of nonstructural components. For this case, it is recommended that amplification be considered using such higher mode periods in lieu of the higher fundamental period. This approach may be generalized by computing floor response spectra for various levels that reflect the dynamic characteristics of the supporting structure to determine how amplification varies as a function of component period. Calculation of floor response spectra is described in Section 13.3.1.1. This procedure can be complex, but a simplified procedure is presented in Section 13.3.1.2. Consideration of nonlinear behavior of the structure greatly complicates the analysis.

In ASCE 7-10, values for Ω_0 were added to Tables 13.5-1 and 13.6-1. Ω_0 is included in the load combinations for the anchorage of nonstructural components to concrete or masonry in cases where the anchorage is deemed nonductile. ACI 318-14, Section 17.2.3.4.3 (d) requires the inclusion of Ω_0 for anchors that do not otherwise satisfy ductility requirements. This factor closely resembles the 1.3 factor and R_p penalty in ASCE 7-05 for nonductile anchorages. The upper value for Ω_0 was originally selected based on similar provisions in prior editions of the IBC and ACI codes. Research at San Diego State University indicates that this value need not exceed 2.

Eq. (13.3-1) represents a trapezoidal distribution of floor accelerations within a structure, varying linearly from the acceleration at the ground (taken as $0.4S_{DS}$) to the acceleration at the roof (taken as $1.2S_{DS}$). The ground acceleration ($0.4S_{DS}$) is intended to be the same acceleration used as design input for the structure itself, including site effects. The roof acceleration is established as three times the input ground acceleration based on examination of recorded in-structure acceleration data for short and moderate-height structures in response to large California earthquakes. Work by Dowell and Johnson (2013) suggests that, for taller structures, the amplification with height may vary significantly because of higher mode effects. Where more information is available, Eq. (13.3-4) permits an alternate determination of the component design forces based on the dynamic properties of the structure. When using a modal analysis procedure, as described in Section 12.9.1, the maximum floor accelerations are used.

Eq. (13.3-3) establishes a minimum seismic design force, F_p , which is consistent with current practice. Eq. (13.3-2) provides a simple maximum value of F_p that prevents multiplication of the individual factors from producing a design force that would be unreasonably high, considering the expected nonlinear response of support and component. Fig. C13.3-2 illustrates the distribution of the specified lateral design forces.

For elements with points of attachment at more than one height, it is recommended that design be based on the average of values of F_p determined individually at each point of attachment (but with the entire component weight, W_p) using Eqs. (13.3-1) through (13.3-3).

Alternatively, for each point of attachment, a force F_p may be determined using Eqs. (13.3-1) through (13.3-3), with the portion of the component weight, W_p , tributary to the point of attachment. For design of the component, the attachment force F_p must be distributed relative to the component's mass distribution over the area used to establish the tributary weight. To illustrate these options, consider a solid exterior nonstructural wall panel, supported top and bottom, for a one-story building with a rigid diaphragm. The values of F_p computed, respectively, for the top and bottom attachments using Eqs. (13.3-1) through (13.3-3) are $0.48S_{DS}I_pW_p$ and $0.30S_{DS}I_pW_p$. In the recommended method, a uniform load is applied to the entire panel based on $0.39S_{DS}I_pW_p$. In the alternative method, a trapezoidal load varying from $0.48S_{DS}I_pW_p$ at the top to $0.30S_{DS}I_pW_p$ at the bottom is applied. Each anchorage force is then determined considering static equilibrium of the complete component subject to all the distributed loads.

Cantilever parapets that are part of a continuous element should be checked separately for parapet forces. The seismic force on any component must be applied at the center of gravity of the component and must be assumed to act in any horizontal direction. Vertical forces on nonstructural components equal to $\pm 0.2S_{DS}W_p$ are specified in Section 13.3.1 and are intended to be applied to all nonstructural components and not just cantilevered elements. Nonstructural concrete or masonry walls laterally supported by flexible diaphragms must be anchored out of plane in accordance with Section 12.11.2.

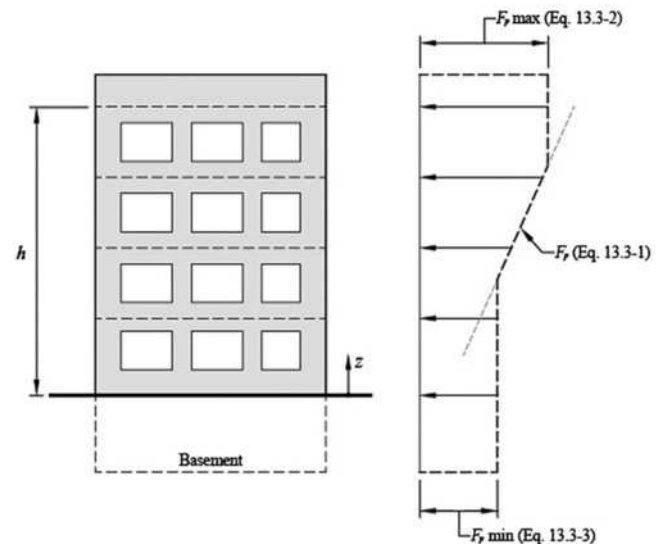


FIGURE C13.3-2 Lateral Force Magnitude over Height

C13.3.1.4 Dynamic Analysis

C13.3.1.4.1 Floor Response Spectra. The response of nonstructural components to earthquake shaking is influenced by the dynamic interaction of the nonstructural component with the response of the structure. Dynamic amplification occurs where the period of a nonstructural component closely matches that of any mode of the supporting structure, although this effect may not be significant depending on the ground motion. For most buildings, the primary mode of vibration in each direction will have the most influence on the dynamic amplification for nonstructural components. For long-period structures (such as tall buildings) higher modes of vibration may have periods that more closely match the period of nonstructural components. For this case, it is recommended that amplification be considered using such higher mode periods in lieu of the higher fundamental period. The approach may be generalized by computing floor response spectra for various levels that reflect the dynamic characteristics of the supporting structure to determine how amplification will vary as a function of component period. In order to properly consider this dynamic amplification, the vibrational characteristics of the building and the component must be known. The characteristics of the building are typically determined from a structural model of the building. The vibrational characteristics of the nonstructural components can be determined either by calculation or by testing. It should be noted that many types of nonstructural components, such as mechanical and electrical equipment, have multiple modes of vibration. The floor response spectra approach may not be applicable to nonstructural components that cannot be characterized as having a predominant mode of vibration.

C13.3.1.4.2 Alternate Floor Response Spectra. Calculation of floor response spectra can be complex since it requires a response history analysis. An alternative method of calculating a floor response history has been presented in Kehoe and Hachem (2003). The procedure described is based on a method of calculating floor response spectra initially developed for the U.S. Army Tri-Service manual (1986). The referenced paper compares the alternate floor response spectra to floor response spectra generated for three example buildings using ground motion records from several California earthquakes.

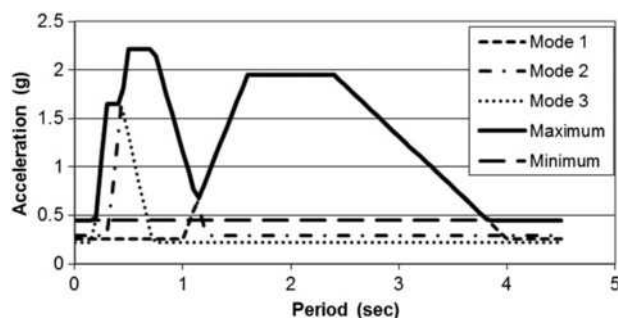


FIGURE C13.3-3 Approximate Floor Response Spectrum

Source: Kehoe and Hachem (2003); reproduced with permission from the Applied Technology Council.

This method considers the dynamic amplification contribution to the nonstructural component response based on the first three modes of vibration of the structure in calculating a floor response spectrum. It is important to note that the use of this procedure requires that the dynamic properties of both the building and the nonstructural components be known. A dynamic amplification factor is applied to each of the modal floor acceleration values for each of the first three modes of vibration, which is calculated based on linear dynamic response spectrum analysis procedures as described in Section 12.9. The dynamic amplification factor is approximated as a function of the ratio of the component period to the period of vibration of a mode of vibration of the building. The magnitude of amplification can vary depending on the ground motion and the building's dynamic characteristics. A peak amplification of 5 has been assumed for the cases where the ratio of component period to building period is between 0.8 and 1.2 using 5% damping for the nonstructural component based on review of results from previous studies (Kehoe and Hachem 2003). Where the ratio of component period to building period is less than 0.5 or more than 2.0, there is no dynamic amplification (Bachman et al. 1993).

The method allows calculation of floor response spectra in each direction at each floor level of the building using the mode shapes and the modal participation factors to amplify the ground motion to each floor level. The method considers linear building response. An example of a floor response spectrum is shown in Fig. C13.3-3. As shown in this figure, the floor response spectrum is taken as the maximum acceleration at each component period from each of the first three modes of the building in each orthogonal direction. The floor response spectrum can be used as a basis for determining the design horizontal force for the nonstructural component by multiplying the acceleration by the component weight and the Importance Factor and dividing by the response factor, R_p .

C13.3.2 Seismic Relative Displacements. The equations of this section are for use in design of cladding, stairways, windows, piping systems, sprinkler components, and other components connected to one structure at multiple levels or to multiple structures. Two equations are given for each situation. Eqs. (13.3-7) and (13.3-9) produce structural displacements as determined by elastic analysis, unreduced by the structural response modification factor (R). Because the actual displacements may not be known when a component is designed or procured, Eqs. (13.3-8) and (13.3-10) provide upper-bound displacements based on structural drift limits. Use of upper-bound equations may facilitate timely design and procurement of components, but may also result in costly added conservatism.

The value of seismic relative displacements is taken as the calculated displacement, D_p , times the Importance Factor, I_e , because the elastic displacement calculated in accordance with Eq. (12.8-15) to establish δ_x (and thus D_p) is adjusted for I_e in keeping with the philosophy of displacement demand for the structure. For component design, the unreduced elastic displacement is appropriate.

The standard does not provide explicit acceptance criteria for the effects of seismic relative displacements, except for glazing. Damage to nonstructural components caused by relative displacement is acceptable, provided that the performance goals defined elsewhere in the chapter are achieved.

The design of some nonstructural components that span vertically in the structure can be complicated when supports for the element do not occur at horizontal diaphragms. The language in Section 13.3.2 was previously amended to clarify that story drift must be accommodated in the elements that actually distort. For example, a glazing system supported by precast concrete spandrels must be designed to accommodate the full story drift, even though the height of the glazing system is only a fraction of the floor-to-floor height. This condition arises because the precast spandrels behave as rigid bodies relative to the glazing system and therefore all the drift must be accommodated by anchorage of the glazing unit, the joint between the precast spandrel and the glazing unit, or some combination of the two.

C13.3.2.1 Displacements within Structures. Seismic relative displacements can subject components or systems to unacceptable stresses. The potential for interaction resulting from component displacements (in particular for distributed systems) and the resulting impact effects should also be considered (see Section 13.2.3).

These interrelationships may govern the clearance requirements between components or between components and the surrounding structure. Where sufficient clearance cannot be provided, consideration should be given to the ductility and strength of the components and associated supports and attachments to accommodate the potential impact.

Where nonstructural components are supported between, rather than at, structural levels, as frequently occurs for glazing systems, partitions, stairs, veneers, and mechanical and electrical distributed systems, the height over which the displacement demand, D_p , must be accommodated may be less than the story height, h_{sx} , and should be considered carefully. For example, consider the glazing system supported by rigid precast concrete spandrels shown in Fig. C13.3-4. The glazing system may be subjected to full story drift, D_p , although its height ($h_x - h_y$) is only a fraction of the story height. The design drift must be accommodated by anchorage of the glazing unit, the joint between the precast spandrel and the glazing unit, or some combination of the two. Similar displacement demands arise where pipes, ducts, or conduits that are braced to the floor or roof above are connected to the top of a tall, rigid, floor-mounted component.

For ductile components, such as steel piping fabricated with welded connections, the relative seismic displacements between support points can be more significant than inertial forces. Ductile piping can accommodate relative displacements by local yielding with strain accumulations well below failure levels. However, for components fabricated using less ductile materials, where local yielding must be avoided to prevent unacceptable failure consequences, relative displacements must be accommodated by flexible connections.

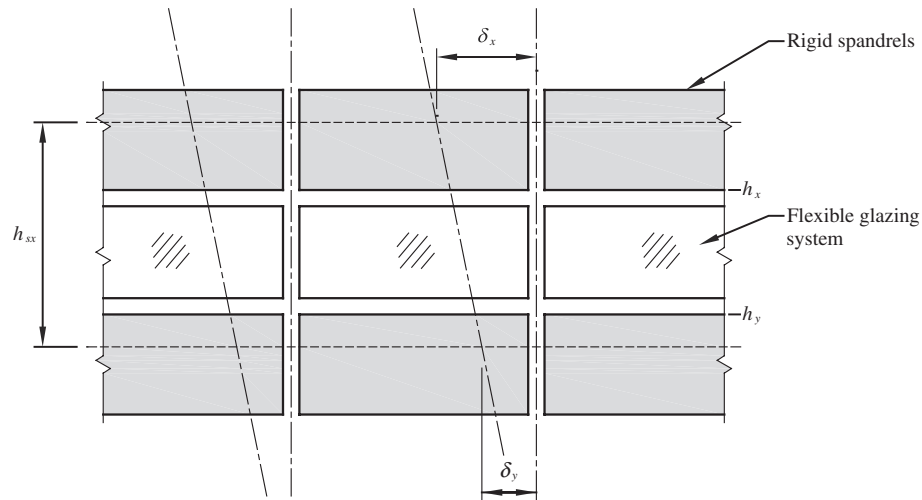


FIGURE C13.3-4 Displacements over Less than Story Height

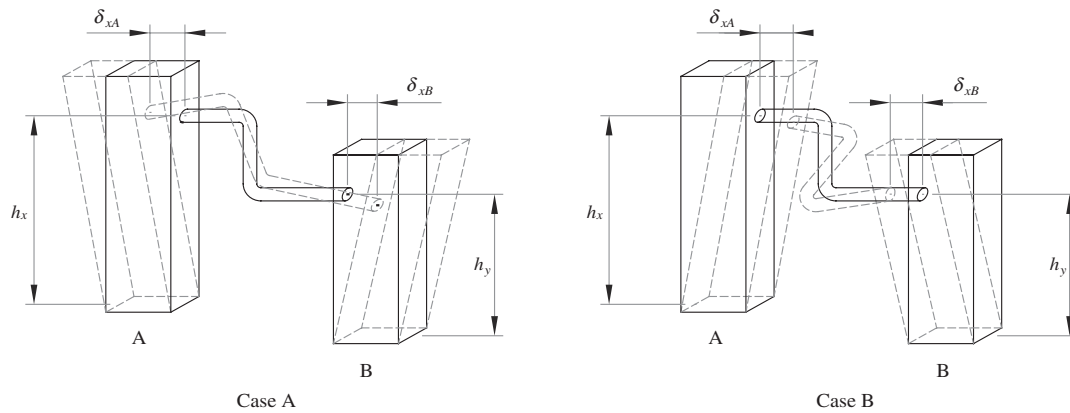


FIGURE C13.3-5 Displacements between Structures

C13.3.2.2 Displacements between Structures. A component or system connected to two structures must accommodate horizontal movements in any direction, as illustrated in Fig. C13.3-5.

C13.3.3 Component Period. Component period is used to classify components as rigid ($T \leq 0.06$ s) or flexible ($T > 0.06$ s). Determination of the fundamental period of an architectural, mechanical, or electrical component using analytical or test methods is often difficult and, if not properly performed, may yield incorrect results. In the case of mechanical and electrical equipment, the flexibility of component supports and attachments typically dominates component response and fundamental component period, and analytical determinations of component period should consider those sources of flexibility. Where testing is used, the dominant mode of vibration of concern for seismic evaluation must be excited and captured by the test setup. The dominant mode of vibration for these types of components cannot generally be acquired through in situ tests that measure only ambient vibrations. To excite the mode of vibration with the highest fundamental period by in situ testing, relatively significant input levels of motion may be required to activate the flexibility of the base and attachment. A resonant frequency search procedure, such as that given in ICC-ES acceptance criteria (AC156 2010), may be used to identify the dominant modes of vibration of a component.

Many mechanical components have fundamental periods below 0.06 s and may be considered rigid. Examples include horizontal pumps, engine generators, motor generators, air compressors, and motor-driven centrifugal blowers. Other types of mechanical equipment, while relatively stiff, have fundamental periods (up to about 0.13 s) that do not permit automatic classification as rigid. Examples include belt-driven and vane axial fans, heaters, air handlers, chillers, boilers, heat exchangers, filters, and evaporators. Where such equipment is mounted on vibration isolators, the fundamental period is substantially increased.

Electrical equipment cabinets can have fundamental periods ranging from 0.06 to 0.3 s, depending upon the supported weight and its distribution, the stiffness of the enclosure assembly, the flexibility of the enclosure base, and the load path through to the attachment points. Tall, narrow motor control centers and switchboards lie at the upper end of this period range. Low- and medium-voltage switchgear, transformers, battery chargers, inverters, instrumentation cabinets, and instrumentation racks usually have fundamental periods ranging from 0.1 to 0.2 s. Braced battery racks, stiffened vertical control panels, bench boards, electrical cabinets with top bracing, and wall-mounted panelboards generally have fundamental periods ranging from 0.06 to 0.1 s.

C13.4 NONSTRUCTURAL COMPONENT ANCHORAGE

Unless exempted in Section 13.1.4 or 13.6.9, components must be anchored to the structure, and all required supports and attachments must be detailed in the construction documents. To satisfy the load path requirement of this section, the detailed information described in Section C13.2.7 must be communicated during the design phase to the registered design professional responsible for the design of the supporting structure. The load path includes housekeeping slabs and curbs, which must be adequately reinforced and positively fastened to the supporting structure. Because the exact magnitude and location of the loads imposed on the structure may not be known until nonstructural components are ordered, the initial design of supporting structural elements should be based on conservative assumptions. The design of the supporting structural elements must be verified once the final magnitude and location of the design loads have been established. The limited exception for ballasted rooftop solar panels meeting the requirements of Section 13.6.12 is intended to accommodate the increasing use of such arrays on roof systems where positive attachment is difficult.

Design documents should provide details with sufficient information so that compliance with these provisions can be verified. Parameters such as a_p , R_p , I_p , S_{DS} , and W_p should be noted. Attachment details may include, as appropriate, dimensions and material properties of the connecting material, weld sizes, bolt sizes and material types for steel-to-steel connections, postinstalled anchor types, diameters, embedments, installation requirements, sheet metal screw diameters and material thicknesses of the connected parts, wood fastener types, and minimum requirements for specific gravity of the base materials.

Seismic design forces are determined using the provisions of Section 13.3.1. Specific reference standards should be consulted for additional adjustments to loads or strengths. Refer, for example, to the anchor design provisions of ACI 318, Chapter 17, for specific provisions related to seismic design of anchors in concrete. Unanchored components often rock or slide when subjected to earthquake motions. Because this behavior may have serious consequences, is difficult to predict, and is exacerbated by vertical ground motions, positive restraint must be provided for each component.

The effective seismic weight used in design of the seismic force-resisting system must include the weight of supported components. To satisfy the load path requirements of this section, localized component demand must also be considered. This satisfaction may be accomplished by checking the capacity of the first structural element in the load path (for example, a floor beam directly under a component) for combined dead, live, operating, and seismic loads, using the horizontal and vertical loads from Section 13.3.1 for the seismic demand, and repeating this procedure for each structural element or connection in the load path until the load case, including horizontal and vertical loads from Section 13.3.1, no longer governs design of the element. The load path includes housekeeping slabs and curbs, which must be adequately reinforced and positively fastened to the supporting structure.

Because the exact magnitude and location of loads imposed on the structure may not be known until nonstructural components are ordered, the initial design of supporting structural elements should be based on conservative assumptions. The design of the supporting structural elements may need to be verified once the final magnitude and location of the design loads have been established.

Tests have shown that there are consistent shear ductility variations between bolts installed in drilled or punched plates

with nuts and connections using welded shear studs. The need for reductions in allowable loads for particular anchor types to account for loss of stiffness and strength may be determined through appropriate dynamic testing. Although comprehensive design recommendations are not available at present, this issue should be considered for critical connections subject to dynamic or seismic loading.

C13.4.1 Design Force in the Attachment. Previous editions of ASCE/SEI 7 included provisions for the amplification of forces to design the component anchorage. These provisions were intended to ensure that the anchorage either (a) would respond to overload in a ductile manner or (b) would be designed so that the anchorage would not be the weakest link in the load path.

Because of the difficulties associated with the application of the anchorage provisions in Section 13.4 in conjunction with anchorage provisions in other reference standards, the provisions for anchorage in ASCE/SEI 7-10 are substantially simplified. Adjustments on the R_p value used for the anchorage calculation have been eliminated, with the exception of the upper limit on R_p of 6, which is intended primarily to address the anchorage of ductile piping systems that are assigned higher R_p values. These higher component response modification factors reflect the inherent ductility and overstrength of ductile piping but may result in an underprediction of the forces on the anchorage.

C13.4.2 Anchors in Concrete or Masonry. Design capacity for anchors in concrete must be determined in accordance with ACI 318, Chapter 17. Design capacity for anchors in masonry is determined in accordance with TMS 402. Anchors must be designed to have ductile behavior or to provide a specified degree of excess strength. Depending on the specifics of the design condition, ductile design of anchors in concrete may satisfy one or more of the following objectives:

1. Adequate load redistribution between anchors in a group;
2. Allowance for anchor overload without brittle failure; or
3. Energy dissipation.

Achieving deformable, energy-absorbing behavior in the anchor itself is often difficult. Unless the design specifically addresses the conditions influencing desirable hysteretic response (e.g., adequate gauge length, anchor spacing, edge distance, and steel properties), anchors cannot be relied on for energy dissipation. Simple geometric rules, such as restrictions on the ratio of anchor embedment length to depth, may not be adequate to produce reliable ductile behavior. For example, a single anchor with sufficient embedment to force ductile tension failure in the steel body of the anchor bolt may still experience concrete fracture (a nonductile failure mode) if the edge distance is small, the anchor is placed in a group of tension-loaded anchors with reduced spacing, or the anchor is loaded in shear instead of tension. In the common case where anchors are subject primarily to shear, response governed by the steel element may be nonductile if the deformation of the anchor is constrained by rigid elements on either side of the joint. Designing the attachment so that its response is governed by a deformable link in the load path to the anchor is encouraged. This approach provides ductility and overstrength in the connection while protecting the anchor from overload. Ductile bolts should only be relied on as the primary ductile mechanism of a system if the bolts are designed to have adequate gauge length (using the unbonded strained length of the bolt) to accommodate the anticipated nonlinear displacements of the system at the design earthquake. Guidance for determining the gauge length can be found in Part 3 of the 2009 NEHRP provisions.

Anchors used to support towers, masts, and equipment are often provided with double nuts for leveling during installation. Where base-plate grout is specified at anchors with double nuts, it should not be relied on to carry loads because it can shrink and crack or be omitted altogether. The design should include the corresponding tension, compression, shear, and flexure loads.

Postinstalled anchors in concrete and masonry should be qualified for seismic loading through appropriate testing. The requisite tests for expansion and undercut anchors in concrete are given in ACI 355.2-07, *Qualification of Post-Installed Mechanical Anchors in Concrete and Commentary* (2007). Testing and assessment procedures based on the ACI standard that address expansion, undercut, screw, and adhesive anchors are incorporated in ICC-ES acceptance criteria. AC193, *Acceptance Criteria for Mechanical Anchors in Concrete Elements* (2012c), and AC308, *Acceptance Criteria for Post-Installed Adhesive Anchors in Concrete Elements* (2012d), refer to ACI 355.4-11, *Qualification of Post-Installed Adhesive Anchors in Concrete and Commentary* (2011c). These criteria, which include specific provisions for screw anchors and adhesive anchors, also reference ACI qualification standards for anchors. For postinstalled anchors in masonry, seismic prequalification procedures are contained in ICC-ES AC01, *Acceptance Criteria for Expansion Anchors in Masonry Elements* (2012b), AC58, *Acceptance Criteria for Adhesive Anchors in Masonry Elements* (2012a), and AC106, *Acceptance Criteria for Predrilled Fasteners (Screw Anchors) in Masonry* (2012e).

Other references to adhesives (such as in Section 13.5.7.2) apply not to adhesive anchors but to steel plates and other structural elements bonded or glued to the surface of another structural component with adhesive; such connections are generally nonductile.

C13.4.3 Installation Conditions. Prying forces on anchors, which result from a lack of rotational stiffness in the connected part, can be critical for anchor design and must be considered explicitly.

For anchorage configurations that do not provide a direct mechanism to transfer compression loads (for example, a base plate that does not bear directly on a slab or deck but is supported on a threaded rod), the design for overturning must reflect the actual stiffness of base plates, equipment, housing, and other elements in the load path when computing the location of the compression centroid and the distribution of uplift loads to the anchors.

C13.4.4 Multiple Attachments. Although the standard does not prohibit the use of single anchor connections, it is good practice to use at least two anchors in any load-carrying connection whose failure might lead to collapse, partial collapse, or disruption of a critical load path.

C13.4.5 Power-Actuated Fasteners. Restrictions on the use of power-actuated fasteners are based on observations of failures of sprinkler pipe runs in the 1994 Northridge earthquake. Although it is unclear from the record to what degree the failures occurred because of poor installation, product deficiency, overload, or consequential damage, the capacity of power-actuated fasteners in concrete often varies more than that of drilled postinstalled anchors. The shallow embedment, small diameter, and friction mechanism of these fasteners make them particularly susceptible to the effects of concrete cracking. The suitability of power-actuated fasteners to resist tension in concrete should be demonstrated by simulated seismic testing in cracked concrete.

Where properly installed in steel, power-actuated fasteners typically exhibit reliable cyclic performance. Nevertheless, they

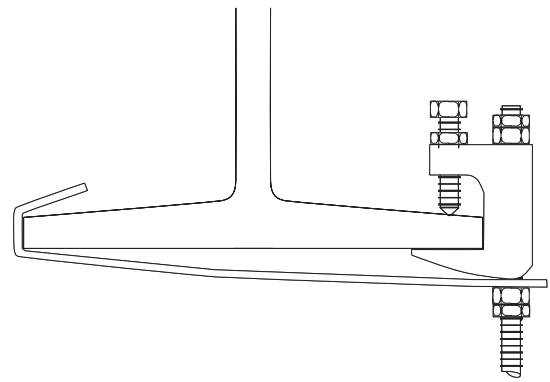


FIGURE C13.4-1 C-Type Beam Clamp Equipped with a Restraining Strap

should not be used singly to support suspended elements. Where used to attach cladding and metal decking, subassembly testing may be used to establish design capacities because the interaction among the decking, the subframe, and the fastener can only be estimated crudely by currently available analysis methods.

The exception permits the use of power-actuated fasteners for specific light-duty applications with upper limits on the load that can be resisted in these cases. All fasteners must have adequate capacity for the calculated loads, including prying forces.

The exception allows for the continued use of power-actuated fasteners in concrete for the vertical support of suspended acoustical tile or lay-in panel ceilings and for other light distributed systems such as small-diameter conduit held to the concrete surface with C-clips. Experience indicates that these applications have performed satisfactorily because of the high degree of redundancy and light loading. Other than ceilings, hung systems should not be included in this exception because of the potential for bending in the fasteners.

The exception for power-actuated fasteners in steel provides a conservative limit on loading. Currently, no accepted procedure exists for the qualification of power-actuated fasteners to resist earthquake loads.

C13.4.6 Friction Clips. The term *friction clip* is defined in Section 11.2 in a general way to encompass C-type beam clamps, as well as cold-formed metal channel (strut) connections. Friction clips are suitable to resist seismic forces provided that they are properly designed and installed, but under no circumstances should they be relied on to resist sustained gravity loads. C-type clamps must be provided with restraining straps, as shown in Fig. C13.4-1.

C13.5 ARCHITECTURAL COMPONENTS

For structures in Risk Categories I through III, the requirements of Section 13.5 are intended to reduce property damage and life-safety hazards posed by architectural components and caused by loss of stability or integrity. When subjected to seismic motion, components may pose a direct falling hazard to building occupants or to people outside the building (as in the case of parapets, exterior cladding, and glazing). Failure or displacement of interior components (such as partitions and ceiling systems in exits and stairwells) may block egress.

For structures in Risk Category IV, the potential disruption of essential function caused by component failure must also be considered.

Architectural component failures in earthquakes can be caused by deficient design or construction of the component,

interrelationship with another component that fails, interaction with the structure, or inadequate attachment or anchorage. For architectural components, attachment and anchorage are typically the most critical concerns related to their seismic performance. Concerns regarding loss of function are most often associated with mechanical and electrical components. Architectural damage, unless severe, can be accommodated temporarily. Severe architectural damage is often accompanied by significant structural damage.

C13.5.1 General. Suspended architectural components are not required to satisfy the force and displacement requirements of Chapter 13, where prescriptive requirements are met. The requirements were relaxed in the 2005 edition of the standard to better reflect the consequences of the expected behavior. For example, impact of a suspended architectural ornament with a sheet metal duct may only dent the duct without causing a credible danger (assuming that the ornament remains intact). The reference to Section 13.2.3 allows the designer to consider such consequences in establishing the design approach.

Nonstructural components supported by chains or otherwise suspended from the structure are exempt from lateral bracing requirements, provided that they are designed not to inflict damage to themselves or any other component when subject to seismic motion. However, for the 2005 edition, it was determined that clarifications were needed on the type of nonstructural components allowed by these exceptions and the acceptable consequences of interaction between components. In ASCE 7-02, certain nonstructural components that could represent a fire hazard after an earthquake were exempted from meeting the Section 9.6.1 requirements. For example, gas-fired space heaters clearly pose a fire hazard after an earthquake, but were permitted to be exempted from the ASCE 7-02 Section 9.6.1 requirements. The fire hazard after the seismic event must be given the same level of consideration as the structural failure hazard when considering components to be covered by this exception. In addition, the ASCE 7-02 language was sometimes overly restrictive because it did not distinguish between credible seismic interactions and incidental interactions. In ASCE 7-02, if a suspended lighting fixture could hit and dent a sheet metal duct, it would have to be braced, although no credible danger is created by the impact. The new reference in Section 13.2.3 of ASCE 7-05 allowed the designer to consider whether the failures of the component and/or the adjacent components are likely to occur if contact is made. These provisions were carried into ASCE 7-10.

C13.5.2 Forces and Displacements. Partitions and interior and exterior glazing must accommodate story drift without failure that will cause a life-safety hazard. Design judgment must be used to assess potential life-safety hazards and the likelihood of life-threatening damage. Special detailing to accommodate drift for typical gypsum board or demountable partitions is unlikely to be cost-effective, and damage to these components poses a low hazard to life safety. Damage in these partitions occurs at low drift levels but is inexpensive to repair.

If they must remain intact after strong ground motion, non-structural fire-resistant enclosures and fire-rated partitions require special detailing that provides isolation from the adjacent or enclosing structure for deformation equivalent to the calculated drift (relative displacement). In-plane differential movement between structure and wall is permitted. Provision must be made for out-of-plane restraint. These requirements are particularly important in steel or concrete moment-frame structures, which experience larger drifts. The problem is less likely to be encountered in stiff structures, such as those with shear walls.

Differential vertical movement between horizontal cantilevers in adjacent stories (such as cantilevered floor slabs) has occurred in past earthquakes. The possibility of such effects should be considered in the design of exterior walls.

C13.5.3 Exterior Nonstructural Wall Elements and Connections. Nonbearing wall panels that are attached to and enclose the structure must be designed to resist seismic (inertial) forces, wind forces, and gravity forces and to accommodate movements of the structure resulting from lateral forces and temperature change. The connections must allow wall panel movements caused by thermal and moisture changes and must be designed to prevent the loss of load-carrying capacity in the event of significant yielding. Where wind loads govern, common practice is to design connectors and panels to allow for not less than two times the story drift caused by wind loads determined, using a return period appropriate to the site location.

Design to accommodate seismic relative displacements often presents a greater challenge than design for strength. Story drifts can amount to 2 in. (50 mm) or more. Separations between adjacent panels are intended to limit contact and resulting panel misalignment or damage under all but extreme building response. Section 13.5.3, item 1, calls for a minimum separation of 1/2 in. (13 mm). For practical joint detailing and acceptable appearance, separations typically are limited to about 3/4 in. (19 mm). Manufacturing and construction tolerances for both wall elements and the supporting structure must be considered in establishing design joint dimensions and connection details.

Cladding elements, which are often stiff in-plane, must be isolated so that they do not restrain and are not loaded by drift of the supporting structure. Slotted connections can provide isolation, but connections with long rods that flex achieve the desired behavior without requiring precise installation. Such rods must be designed to resist tension and compression in addition to induced flexural stresses and brittle, low-cycle fatigue failure.

Full-story wall panels are usually rigidly attached to and move with the floor structure nearest the panel bottom and isolated at the upper attachments. Panels also can be vertically supported at the top connections with isolation connections at the bottom. An advantage of this configuration is that failure of an isolation connection is less likely to result in complete detachment of the panel because it tends to rotate into the structure rather than away from it.

To minimize the effects of thermal movements and shrinkage on architectural cladding panels, connection systems are generally detailed to be statically determinate. Because the resulting support systems often lack redundancy, exacerbating the consequences of a single connection failure, fasteners must be designed for amplified forces and connecting members must be ductile. The intent is to keep inelastic behavior in the connecting members while the more brittle fasteners remain essentially elastic. To achieve this intent, the tabulated a_p , R_p , and Ω_0 values produce fastener design forces that are about three times those for the connecting members.

Limited deformability curtain walls, such as aluminum systems, are generally light and can undergo large deformations without separating from the structure. However, care must be taken in design of these elements so that low deformability components (as defined in Section 11.2) that may be part of the system, such as glazing panels, are detailed to accommodate the expected deformations without failure.

In Table 13.5-1, veneers are classified as either limited or low-deformability elements. Veneers with limited deformability, such as vinyl siding, pose little risk. Veneers with low deformability, such as brick and ceramic tile, are highly sensitive to the

performance of the supporting substrate. Significant distortion of the substrate results in veneer damage, possibly including separation from the structure. The resulting risk depends on the size and weight of fragments likely to be dislodged and on the height from which the fragments would fall. Detachment of large portions of the veneer can pose a significant risk to life. Such damage can be reduced by isolating veneer from displacements of the supporting structure. For structures with flexible lateral force-resisting systems, such as moment frames and buckling-restrained braced frames, approaches used to design nonbearing wall panels to accommodate story drift should be applied to veneers.

The limits on length to diameter ratios are needed to ensure proper connection performance. Recent full-scale building shake table tests conducted at University of California, San Diego, demonstrated that sliding connections perform well when the rod is short. Longer rods in sliding connections bind if there is significant bending and rotation in the rod, which may lead to a brittle failure. For rods that accommodate drift by flexure, longer rods reduce inelastic bending demands and provide better performance. Since anchor rods used in sliding and bending may undergo inelastic action, the use of mild steel improves ductility.

Threaded rods subjected to bending have natural notches (the threads) and are therefore a concern for fatigue. In high-seismic applications, the response may induce a high bending demand and low-cycle fatigue. Cold-worked threaded rod offers significantly reduced ductility unless annealed. Rods meeting the requirements of ASTM F1554, Grade 36, in their as-fabricated condition (i.e., after threading) provide the desired level of performance. ASTM 1554 rods that fulfill the requirements of Supplement 1 for Grade 55 Bars and Anchor Bolts are also acceptable. Other grades that may also be acceptable include ASTM A36, A307, A572, Grade 50, and A588. Other connection configurations and materials may be used, provided that they are approved in accordance with ASCE 7-16 Section 1.3.1.3 and are designed to accommodate the story drift without brittle failure.

The reference to D_p has been changed to D_{pl} to reflect consideration of the earthquake Importance Factor on drift demands. Connections should include a means for accommodating erection tolerance so that the required connection capacity is maintained.

C13.5.4 Glass. Glass is commonly secured to the window system framing by a glazing pocket built into the framing. This is commonly referred to as a *mechanically captured* or *dry-glazed* window system. Glass can also be secured to the window system framing with a structural silicone sealant. This is commonly referred to as a *wet-glazed* window system. Imposed loads are transferred from the glass to the window system framing through the adhesive bond of the structural silicone sealant. ASTM C1401 *Standard Guide for Structural Sealant Glazing* (2014b) provides guidance and reference standards for manufacture, testing, design and installation of structural silicone sealant. This standard addresses glazing sloped to a maximum of 15° from vertical. For glazing slopes exceeding 15°, additional general building code requirements pertaining to sloped glazing and skylights apply.

C13.5.5 Out-of-Plane Bending. The effects of out-of-plane application of seismic forces (defined in Section 13.3.1) on nonstructural walls, including the resulting deformations, must be considered. Where weak or brittle materials are used, conventional deflection limits are expressed as a proportion of the span. The intent is to preclude out-of-plane failure of heavy materials (such as brick or block) or applied finishes (such as stone or tile).

C13.5.6 Suspended Ceilings. Suspended ceiling systems are fabricated using a wide range of building materials with differing

characteristics. Some systems (such as gypsum board, screwed or nailed to suspended members) are fairly homogeneous and should be designed as light-frame diaphragm assemblies, using the forces of Section 13.3 and the applicable material-specific design provisions of Chapter 14. Others are composed of discrete elements laid into a suspension system and are the subject of this section.

Seismic performance of ceiling systems with lay-in or acoustical panels depends on support of the grid and individual panels at walls and expansion joints, integrity of the grid and panel assembly, interaction with other systems (such as fire sprinklers), and support for other nonstructural components (such as light fixtures and HVACR systems). Observed performance problems include dislodgement of tiles because of impact with walls and water damage (sometimes leading to loss of occupancy) because of interaction with fire sprinklers.

Suspended lath and plaster ceilings are not exempted from the requirements of this section because of their more significant mass and the greater potential for harm associated with their failure. However, the prescriptive seismic provisions of Section 13.5.6.2 and ASTM E580 for acoustical tile and lay-in panel ceilings, including the use of compression posts, are not directly applicable to these systems primarily because of their behavior as a continuous diaphragm and greater mass. As such, they require more attention to design and detailing, in particular for the attachment of the hanger wires to the structure and main carriers, the attachment of the cross-furring channels to main carriers, and the attachment of lath to cross-furring channels. Attention should also be given to the attachment of light fixtures and diffusers to the ceiling structure. Bracing should consider both horizontal and vertical movement of the ceiling, as well as discontinuities and offsets. The seismic design and detailing of lath and plaster ceilings should use rational engineering methods to transfer seismic design ceiling forces to the building structural elements.

The performance of ceiling systems is affected by the placement of seismic bracing and the layout of light fixtures and other supported loads. Dynamic testing has demonstrated that splayed wires, even with vertical compression struts, may not adequately limit lateral motion of the ceiling system caused by straightening of the end loops. Construction problems include slack installation or omission of bracing wires caused by obstructions. Other testing has shown that unbraced systems may perform well where the system can accommodate the expected displacements, by providing both sufficient clearance at penetrations and wide closure members, which are now required by the standard.

With reference to the exceptions in Section 13.5.6,

- The first exemption is based on the presumption that lateral support is accomplished by the surrounding walls for areas equal to or less than 144 ft² (13.4 m²) (e.g., a 12-ft by 12-ft (3.7-m by 3.7-m) room). The 144 ft² (13.4 m²) limit corresponds historically to an assumed connection strength of 180 lb (4.5 N) and forces associated with requirements for suspended ceilings that first appeared in the 1976 Uniform Building Code.
- The second exemption assumes that planar, horizontal drywall ceilings behave as diaphragms (i.e., develop in-plane strength). This assumption is supported by the performance of drywall ceilings in past earthquakes.

C13.5.6.1 Seismic Forces. Where the weight of the ceiling system is distributed nonuniformly, that condition should be considered in the design because the typical T-bar ceiling grid has limited ability to redistribute lateral loads.

Table C13.5-1 Summary of Requirements for Acoustical Tile or Lay-in Panel Ceilings

Item	Seismic Design Category C	Seismic Design Categories D, E, and F
	Less Than or Equal to 144 ft²	
NA	No requirements. (§1.4)	No requirements. (§1.4)
	Greater than 144 ft² but less than or equal to 1,000 ft²	
Duty Rating	Only Intermediate or Heavy Duty Load Rated grid as defined by ASTM C635 may be used for commercial ceilings. (ASTM C635 sections 4.1.3.1, 4.1.3.2, & 4.1.3.3)	Heavy Duty Load Rating as defined in ASTM C635 is required. (§5.1.1)
Grid Connections	Minimum main tee connection and cross tee intersection strength of 60 lb. (§4.1.2)	Minimum main tee connection and cross tee intersection strength of 180 lb. (§5.1.2)
Vertical Suspension Wires	Vertical hanger wires must be a minimum of 12 gauge. (§4.3.1) Vertical hanger wires maximum 4 ft on center. (§4.3.1) Vertical hanger wires must be sharply bent and wrapped with three turns in 3 in. or less. (§4.3.2) All vertical hanger wires may not be more than 1 in 6 out of plumb without having additional wires counter-splayed. (§4.3.3) Any connection device from the vertical hanger wire to the structure above must sustain a minimum load of 90 lb. (§4.3.2)	Vertical hanger wire must be a minimum of 12 gauge. (§5.2.7.1) Vertical hanger wires maximum 4 ft on center. (§5.2.7.1) Vertical hanger wires must be sharply bent and wrapped with three turns in 3 in. or less. (§5.2.7.2) All vertical hanger wires may not be more than 1 in 6 out of plumb without having additional wires counter-splayed. (§5.2.7.3) Any connection device from the vertical hanger wire to the structure above must sustain a minimum load of 90 lb. (§5.2.7.2)
Lateral Bracing	Wires may not attach to or bend around interfering equipment without the use of trapezes. (§4.3.4) Lateral bracing is not permitted. Ceiling is intended to “float” relative to balance of structure. Tee connections may be insufficient to maintain integrity if braces were included. NOTE 1)	Wires may not attach to or bend around interfering equipment without the use of trapezes. (§5.2.7.4) Not required under 1,000 ft ² . For ceiling areas under 1,000 ft ² , perimeter and tee connections are presumed to be sufficiently strong to maintain integrity whether bracing is installed or not. (§5.2.8.1)
Perimeter	Perimeter closure (molding) width must be a minimum of 7/8 in. (§4.2.2) Perimeter closures with a support ledge of less than 7/8-in. shall be supported by perimeter vertical hanger wires not more than 8 in. from the wall. (§4.2.3) A minimum clearance of 3/8 in. must be maintained on all four sides. (§4.2.4) Permanent attachment of grid ends is not permitted (§4.2.6)	Perimeter closure (molding) width must be a minimum of 2 in. (§5.2.2) Two adjacent sides must be connected to the wall or perimeter closure. (§5.2.3) A minimum clearance of 3/4 in. must be maintained on the other two adjacent sides. (§5.2.3) Perimeter tees must be supported by vertical hanger wires not more than 8 in. from the wall. (§5.2.6)
Light Fixtures	Perimeter tee ends must be prevented from spreading. (§4.2.5) Light fixtures must be positively attached to the grid by at least two connections, each capable of supporting the weight of the lighting fixture. (§4.4.1 and NEC) Surface-mounted light fixtures shall be positively clamped to the grid. (§4.4.2) Clamping devices for surface-mounted light fixtures shall have safety wires to the ceiling hanger or to the structure above. (§4.4.2) Light fixtures and attachments weighing 10 lb or less require one number 12 gauge (minimum) hanger wire connected to the housing (e.g., canister light fixture). This wire may be slack. (§4.4.3) Light fixtures that weigh greater than 10 but less than or equal to 56 lb require two number 12 gauge (minimum) hanger wires connected to the housing. These wires may be slack. (§4.4.4) Light fixtures that weigh more than 56 lb require independent support from the structure. (§4.4.5) Pendant-hung light fixtures shall be supported by a minimum 9-gauge wire or other approved alternate. (§4.4.6) Rigid conduit is not permitted for the attachment of fixtures. (§4.4.7)	Perimeter tee ends must be prevented from spreading. (§5.2.4) Light fixtures must be positively attached to the grid by at least two connections, each capable of supporting the weight of the lighting fixture. (NEC, §5.3.1) Surface-mounted light fixtures shall be positively clamped to the grid. (§5.3.2) Clamping devices for surface-mounted light fixtures shall have safety wires to the ceiling wire or to the structure above. (§5.3.2) When cross tees with a load-carrying capacity of less than 16 lb/ft are used, supplementary hanger wires are required. (§5.3.3) Light fixtures and attachments weighing 10 lb or less require one 12-gauge minimum hanger wire connected to the housing and connected to the structure above. This wire may be slack. (§5.3.4) Light fixtures that weigh greater than 10 but less than or equal to 56 lb require two number 12 gauge minimum hanger wires attached to the fixture housing and connected to the structure above. These wires may be slack. (§5.3.5) Light fixtures that weigh more than 56 lb require independent support from the structure by approved hangers. (§5.3.6) Pendant-hung light fixtures shall be supported by a minimum 9-gauge wire or other approved support. (§5.3.7) Rigid conduit is not permitted for the attachment of fixtures. (§5.3.8)

Table C13.5-1 (Continued) Summary of Requirements for Acoustical Tile or Lay-in Panel Ceilings

Item	Seismic Design Category C	Seismic Design Categories D, E, and F
Mechanical Services	Flexibly mounted mechanical services weighing less than or equal to 20 lb must be positively attached to main runners or cross runners with the same load-carrying capacity as the main runners. (§4.5.1) Flexibly mounted mechanical services weighing more than 20 lb but less than or equal to 56 lb must be positively attached to main runners or cross runners with the same load-carrying capacity as the main runners and require two 12-gauge (minimum) hanger wires. These wires may be slack. (§4.5.2) Flexibly mounted mechanical services greater than 56 lb require direct support from the structure. (§4.5.3)	Flexibly mounted mechanical services weighing less than or equal to 20 lb must be positively attached to main runners or cross runners with the same load-carrying capacity as the main runners. (§5.4.1) Flexibly mounted mechanical services weighing more than 20 lb but less than or equal to 56 lb must be positively attached to main runners or cross runners with the same load-carrying capacity as the main runners and require two 12-gauge (minimum) hanger wires. These wires may be slack. (§5.4.2) Flexibly mounted mechanical services greater than 56 lb require direct support from the structure. (§5.4.3)
Supplemental Requirements	All ceiling penetrations must have a minimum of 3/8 in. clearance on all sides. (§4.2.4)	Direct concealed systems must have stabilizer bars or mechanically connected cross tees a maximum of 60 in. on center with stabilization within 24 in. of the perimeter. (§5.2.5) Bracing is required for ceiling plane elevation changes. (§5.2.8.6) Cable trays and electrical conduits shall be supported independently of the ceiling. (§5.2.8.7) 2,500 ft² All ceiling penetrations and independently supported fixtures or services must have closures that allow for a 1-in. movement. (§5.2.8.5) An integral ceiling sprinkler system may be designed by the licensed design professional to eliminate the required spacing of penetrations. (§5.2.8.8) A licensed design professional must review the interaction of nonessential ceiling components with essential ceiling components to prevent the failure of the essential components. (§5.7.1)
Partitions	The ceiling may not provide lateral support to partitions. (§4.6.1) Partitions attached to the ceiling must use flexible connections to avoid transferring force to the ceiling. (§4.6.1)	Partition attached to the ceiling and all partitions greater than feet in height shall be laterally braced to the building structure. This bracing must be independent of the ceiling. (§5.5.1)
Exceptions	The ceiling weight must be less than 2.5 lb=ft², otherwise the prescribed construction for Seismic Design Categories D, E, and F must be used. (§4.1.1)	None.
Lateral Bracing	No additional requirements.	Greater than 1,000 ft² but less than or equal to 2,500 ft² Lateral force bracing (4, 12 gauge splay wires) is required within 2 in. of main tee/cross tee intersection and splayed 90 deg apart in the plan view, at maximum 45-deg angle from the horizontal and located 12 ft on center in both directions, starting 6 ft from walls. (§5.2.8.1 & §5.2.8.2) Lateral force bracing must be spaced a minimum of 6 in. from unbraced horizontal piping or ductwork. (§5.2.8.3) Lateral force bracing connection strength must be a minimum of 250 lb. (§5.2.8.3) Rigid bracing designed to limit deflection at the point of attachment to less than 0.25 in. may be used in place of splay wires. Unless rigid bracing is used or calculations have shown that lateral deflection is less than 1/4 in., sprinkler heads and other penetrations shall have a minimum of 1-in. clear space in all directions. (§5.2.8.5)
Special Considerations	No additional requirements.	Greater than 2,500 ft² Seismic separation joints with a minimum or 3/4-in. axial movement, bulkhead, or full-height partitions with the usual 2-in. closure and other requirements. (§5.2.9.1) Areas defined by seismic separation joints, bulkheads, or full-height partitions must have a ratio of long to short dimensions of less than or equal to 4. (§5.2.9.1)

Notes: There are no requirements for suspended ceilings located in structures assigned to Seismic Design Categories A and B. Unless otherwise noted, all section references in parentheses (§) refer to ASTM E580 (2014).

C13.5.6.2 Industry Standard Construction for Acoustical Tile or Lay-In Panel Ceilings. The key to good seismic performance is sufficiently wide closure angles at the perimeter to accommodate relative ceiling motion and adequate clearance at penetrating components (such as columns and piping) to avoid concentrating restraining loads on the ceiling system.

Table C13.5-1 provides an overview of the combined requirements of ASCE/SEI 7 and ASTM E580 (2014a). Careful review of both documents is required to determine the actual requirements.

C13.5.6.2.1 Seismic Design Category C. The prescribed method for SDC C is a floating ceiling. The design assumes a small displacement of the building structure caused by the earthquake at the ceiling and isolates the ceiling from the perimeter. The vertical hanger wires are not capable of transmitting significant movement or horizontal force into the ceiling system, and therefore the ceiling does not experience significant force or displacement as long as the perimeter gap is not exceeded. All penetrations and services must be isolated from the building structure for this construction method to be effective. If this isolation is impractical or undesirable, the prescribed construction for SDCs D, E, and F may be used.

C13.5.6.2.2 Seismic Design Categories D through F. The industry standard construction addressed in this section relies on ceiling contact with the perimeter wall for restraint.

Typical splay wire lateral bracing allows for some movement before it effectively restrains the ceiling. The intent of the 2-in. (50-mm) perimeter closure wall angle is to permit back-and-forth motion of the ceiling during an earthquake without loss of support, and the width of the closure angle is important to good performance. This standard has been experimentally verified by large-scale testing conducted by ANCO Engineers, Inc., in 1983.

Extensive shake table testing using the protocol contained in ICC-ES AC156 by major manufacturers of suspended ceilings has been used to justify the use of perimeter clips designed to accommodate the same degree of movement as the closure angle while supporting the tee ends. These tests are conducted on 16-ft by 16-ft (4.9-m by 4.9-m) ceiling installations. Testing on larger ceiling systems reported by Rahmanishamsi et al. (2014) and Soroushian et al. (2012, 2014) indicates that the use of approved perimeter clips may lead to damage to the grid members and seismic clips, crushing of wall angles, and deformation of grid latches at moderate ground motion levels if the grid member loses contact with the horizontal leg of the closure angle or channel. A requirement has been added to screw the clips to the closure angle or channel to prevent this type of damage.

The requirement for a 1-in. (25-mm) clearance around sprinkler drops found in Section 13.5.6.2.2 (e) of ASCE/SEI 7-05 is maintained and is contained in ASTM E580.

This seismic separation joint is intended to break the ceiling into isolated areas, preventing large-scale force transfer across the ceiling. The new requirement to accommodate 3/4-in. (19-mm) axial movement specifies the performance requirement for the separation joint.

The requirement for seismic separation joints to limit ceiling areas to 2,500 ft² (232.3 m²) is intended to prevent overload of the connections to the perimeter angle. Limiting the ratio of long to short dimensions to 4:1 prevents dividing the ceiling into long and narrow sections, which could defeat the purpose of the separation.

C13.5.6.3 Integral Construction. Ceiling systems that use integral construction are constructed of modular pre-engineered

components that integrate lights, ventilation components, fire sprinklers, and seismic bracing into a complete system. They may include aluminum, steel, and PVC components and may be designed using integral construction of ceiling and wall. They often use rigid grid and bracing systems, which provide lateral support for all the ceiling components, including sprinkler drops. This bracing reduces the potential for adverse interactions among components and eliminates the need to provide clearances for differential movement.

C13.5.7 Access Floors

C13.5.7.1 General. In past earthquakes and in cyclic load tests, some typical raised access floor systems behaved in a brittle manner and exhibited little reserve capacity beyond initial yielding or failure of critical connections. Testing shows that unrestrained individual floor panels may pop out of the supporting grid unless they are mechanically fastened to supporting pedestals or stringers. This fault may be a concern, particularly in egress pathways.

For systems with floor stringers, it is accepted practice to calculate the seismic force, F_p , for the entire access floor system within a partitioned space and then distribute the total force to the individual braces or pedestals. For stringerless systems, the seismic load path should be established explicitly.

Overtuning effects subject individual pedestals to vertical loads well in excess of the weight, W_p , used in determining the seismic force, F_p . It is unconservative to use the design vertical load simultaneously with the design seismic force for design of anchor bolts, pedestal bending, and pedestal welds to base plates. “Slip-on” heads that are not mechanically fastened to the pedestal shaft and thus cannot transfer tension are likely unable to transfer to the pedestal the overturning moments generated by equipment attached to adjacent floor panels.

To preclude brittle failure, each element in the seismic load path must have energy-absorbing capacity. Buckling failure modes should be prevented. Lower seismic force demands are allowed for special access floors that are designed to preclude brittle and buckling failure modes.

C13.5.7.2 Special Access Floors. An access floor can be a “special access floor” if the registered design professional opts to comply with the requirements of Section 13.5.7.2. Special access floors include construction features that improve the performance and reliability of the floor system under seismic loading. The provisions focus on providing an engineered load path for seismic shear and overturning forces. Special access floors are designed for smaller lateral forces, and their use is encouraged at facilities with higher nonstructural performance objectives.

C13.5.8 Partitions. Partitions subject to these requirements must have independent lateral support bracing from the top of the partition to the building structure or to a substructure attached to the building structure. Some partitions are designed to span vertically from the floor to a suspended ceiling system. The ceiling system must be designed to provide lateral support for the top of the partition. An exception to this condition is provided to exempt bracing of light (gypsum board) partitions where the load does not exceed the minimum partition lateral load. Experience has shown that partitions subjected to the minimum load can be braced to the ceiling without failure.

C13.5.9 Glass in Glazed Curtain Walls, Glazed Storefronts, and Glazed Partitions. The performance of glass in earthquakes falls into one of four categories:

1. The glass remains unbroken in its frame or anchorage.

2. The glass cracks but remains in its frame or anchorage while continuing to provide a weather barrier and to be otherwise serviceable.
3. The glass shatters but remains in its frame or anchorage in a precarious condition, likely to fall out at any time.
4. The glass falls out of its frame or anchorage, either in shards or as whole panels.

Categories 1 and 2 satisfy both Immediate Occupancy and Life Safety Performance Objectives. Although the glass is cracked in Category 2, immediate replacement is not required. Categories 3 and 4 cannot provide for immediate occupancy, and their provision of life safety depends on the post-breakage characteristics of the glass and the height from which it can fall. Tempered glass shatters into multiple, pebble-size fragments that fall from the frame or anchorage in clusters. These broken glass clusters are relatively harmless to humans when they fall from limited heights, but they could be harmful when they fall from greater heights.

C13.5.9.1 General. Eq. (13.5-2) is derived from Sheet Glass Association of Japan (1982) and is similar to an equation in Bouwkamp and Meehan (1960) that permits calculation of the story drift required to cause glass-to-frame contact in a given rectangular window frame. Both calculations are based on the principle that a rectangular window frame (specifically, one that is anchored mechanically to adjacent stories of a structure) becomes a parallelogram as a result of story drift, and that glass-to-frame contact occurs when the length of the shorter diagonal of the parallelogram is equal to the diagonal of the glass panel itself. The value Δ_{fallout} represents the displacement capacity of the system, and D_p represents the displacement demand.

The 1.25 factor in the requirements described above reflects uncertainties associated with calculated inelastic seismic displacements of building structures. Wright (1989) states that post-elastic deformations, calculated using the structural analysis process may well underestimate the actual building deformation by up to 30%. It would therefore be reasonable to require the curtain wall glazing system to withstand 1.25 times the computed maximum interstory displacement to verify adequate performance.

The reason for the second exception to Eq. (13.5-2) is that the tempered glass, if shattered, would not produce an overhead falling hazard to adjacent pedestrians, although some pieces of glass may fall out of the frame.

C13.5.9.2 Seismic Drift Limits for Glass Components. As an alternative to the prescriptive approach of Section 13.5.9.1, the deformation capacity of glazed curtain wall systems may be established by test.

C13.5.10 Egress Stairs and Ramps. In the Christchurch earthquake of February 22, 2011, several buildings using precast concrete stairs provided with a sliding joint at one end experienced stair collapse (Canterbury Earthquakes Royal Commission 2012). In one notable case, the 18-story Forsyth Barr office building, the structure was otherwise largely undamaged. In all cases, the primary cause of collapse was loss of vertical bearing at the end connection due to building drift that exceeded the support detail capacity. These stairs, in general, were intended to serve as egress routes, and occupants were trapped in some of these buildings following the earthquake. In U.S. practice, precast stairs (Fig. C13.5-1) are less common than steel-framed stairs (Fig. C13.5-2), which are generally considered to be inherently flexible. But in shake table tests conducted at the University of California, San Diego, as part of the Network for Earthquake Engineering Simulation (NEES) project, “Full-Scale Structural and Nonstructural Building System Performance during Earthquakes

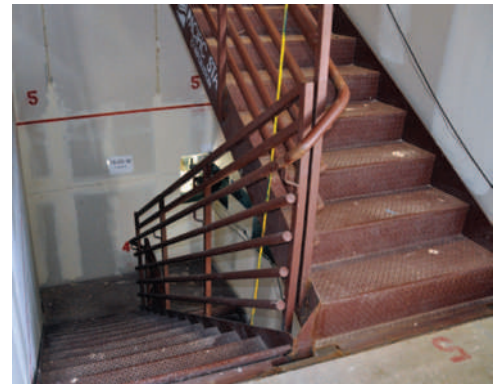


FIGURE C13.5-1 Precast Stair

Source: Courtesy of Tindall Corp.



FIGURE C13.5-2 Steel-Framed Exit Stair

Source: Courtesy of Tara Hutchinson.

and Post-Earthquake Fire,” connections of the commercial metal stair included in the test structure were shown to be brittle and susceptible to damage. Considering the critical nature of egress for life safety, specific attention to the ability of egress stairs to accept building drift demands is warranted. Effective sliding joints in typical steel stairs are complex to design and construct. Ductile connections, capable of accepting the drift without loss of vertical load-carrying capacity are often preferred. In such cases, sufficient ductility must be provided in these connections to accommodate multiple cycles at anticipated maximum drift levels. If drift is to be accommodated with full sliding connections lacking a fail-safe stop, additional length of bearing is required to prevent collapse where displacements exceed design levels. Where stair systems are rigidly attached to the structure, they must be included in the structure model, and the resultant forces must be accommodated, with overstrength, in the stair design.

These requirements do not apply to egress stair systems and ramps that are integral with the building structure since it is assumed that the seismic resistance of these systems is addressed in the overall building design. Examples include stairs and ramps comprising monolithic concrete construction, light-frame wood

and cold-formed metal stair systems in multiunit residential construction, and integrally constructed masonry stairs.

C13.6 MECHANICAL AND ELECTRICAL COMPONENTS

These requirements, focused on design of supports and attachments, are intended to reduce the hazard to life posed by loss of component structural stability or integrity. The requirements increase the reliability of component operation but do not address functionality directly. For critical components where operability is vital, Section 13.2.2 provides methods for seismically qualifying the component.

Traditionally, mechanical components (such as tanks and heat exchangers) without rotating or reciprocating components are directly anchored to the structure. Mechanical and electrical equipment components with rotating or reciprocating elements are often isolated from the structure by vibration isolators (using rubber acting in shear, springs, or air cushions). Heavy mechanical equipment (such as large boilers) may not be restrained at all, and electrical equipment other than generators, which are normally isolated to dampen vibrations, usually is rigidly anchored (for example, switch gear and motor control centers).

Two distinct levels of earthquake safety are considered in the design of mechanical and electrical components. At the usual safety level, failure of the mechanical or electrical component itself because of seismic effects poses no significant hazard. In this case, design of the supports and attachments to the structure is required to avoid a life-safety hazard. At the higher safety level, the component must continue to function acceptably after the design earthquake. Such components are defined as designated seismic systems in Section 11.2 and may be required to meet the special certification requirements of Section 13.2.2.

Not all equipment or parts of equipment need to be designed for seismic forces. Where I_p is specified to be 1.0, damage to, or even failure of, a piece or part of a component does not violate these requirements as long as a life-safety hazard is not created. The restraint or containment of a falling, breaking, or toppling component (or its parts) by means of bumpers, braces, guys, wedges, shims, tethers, or gapped restraints to satisfy these requirements often is acceptable, although the component itself may suffer damage.

Judgment is required to fulfill the intent of these requirements; the key consideration is the threat to life safety. For example, a nonessential air handler package unit that is less than 4 ft (1.2 m) tall bolted to a mechanical room floor is not a threat to life as long as it is prevented from significant displacement by having adequate anchorage. In this case, seismic design of the air handler itself is unnecessary. However, a 10-ft (3.0-m) tall tank on 6-ft (1.8-m) long angles used as legs, mounted on a roof near a building exit does pose a hazard. The intent of these requirements is that the supports and attachments (tank legs, connections between the roof and the legs, and connections between the legs and the tank), and possibly even the tank itself be designed to resist seismic forces. Alternatively, restraint of the tank by guys or bracing could be acceptable.

It is not the intent of the standard to require the seismic design of shafts, buckets, cranks, pistons, plungers, impellers, rotors, stators, bearings, switches, gears, nonpressure retaining casings and castings, or similar items. Where the potential for a hazard to life exists, the design effort should focus on equipment supports, including base plates, anchorages, support lugs, legs, feet, saddles, skirts, hangers, braces, and ties.

Many mechanical and electrical components consist of complex assemblies of parts that are manufactured in an industrial

process that produces similar or identical items. Such equipment may include manufacturers' catalog items and often are designed by empirical (trial-and-error) means for functional and transportation loadings. A characteristic of such equipment is that it may be inherently rugged. The term "rugged" refers to an amplex of construction that provides such equipment with the ability to survive strong motions without significant loss of function. By examining such equipment, an experienced design professional usually should be able to confirm such ruggedness. The results of an assessment of equipment ruggedness may be used in determining an appropriate method and extent of seismic design or qualification effort.

The revisions to Table 13.6-1 in ASCE/SEI 07-10 were the result of work done in recent years to better understand the performance of mechanical and electrical components and their attachment to the structure. The primary concepts of flexible and rigid equipment and ductile and rugged behavior are drawn from SEAOC (1999), Commentary Section C107.1.7. Material on HVACR is based on ASHRAE (2000). Other material on industrial piping, boilers, and pressure vessels is based on the American Society of Mechanical Engineers codes and standards publications (ASME 2007, 2010a, 2010b).

C13.6.1 General. The exception allowing unbraced suspended components has been clarified, addressing concerns about the type of nonstructural components allowed by these exceptions, as well as the acceptable consequences of interaction between components. In previous editions of the standard, certain nonstructural components that could represent a fire hazard after an earthquake were exempt from lateral bracing requirements. In the revised exception, reference to Section 13.2.3 addresses such concerns while distinguishing between credible seismic interactions and incidental interactions.

The seismic demand requirements are based on component structural attributes of flexibility (or rigidity) and ruggedness. Table 13.6-1 provides seismic coefficients based on judgments of the component flexibility, expressed in the a_p term, and ruggedness, expressed in the R_p term. It may also be necessary to consider the flexibility and ductility of the attachment system that provides seismic restraint.

Entries for components and systems in Table 13.6-1 are grouped and described to improve clarity of application. Components are divided into three broad groups, within which they are further classified depending on the type of construction or expected seismic behavior. For example, mechanical components include "air-side" components (such as fans and air handlers) that experience dynamic amplification but are light and deformable; "wet-side" components that generally contain liquids (such as boilers and chillers) that are more rigid and somewhat ductile; and rugged components (such as engines, turbines, and pumps) that are of massive construction because of demanding operating loads and that generally perform well in earthquakes, if adequately anchored.

A distinction is made between components isolated using neoprene and those that are spring isolated. Spring-isolated components are assigned a lower R_p value because they tend to have less effective damping. Internally isolated components are classified explicitly to avoid confusion.

C13.6.2 Mechanical Components and C13.6.3 Electrical Components. Most mechanical and electrical equipment is inherently rugged and, where properly attached to the structure, has performed well in past earthquakes. Because the operational and transportation loads for which the equipment is designed typically are larger than those caused by earthquakes, these requirements focus primarily on equipment anchorage and

attachments. However, designated seismic systems, which are required to function after an earthquake or which must maintain containment of flammable or hazardous materials, must themselves be designed for seismic forces or be qualified for seismic loading in accordance with Section 13.2.2.

The likelihood of post-earthquake operability can be increased where the following measures are taken:

1. Internal assemblies, subassemblies, and electrical contacts are attached sufficiently to prevent their being subjected to differential movement or impact with other internal assemblies or the equipment enclosure.
2. Operators, motors, generators, and other such components that are functionally attached to mechanical equipment by means of an operating shaft or mechanism are structurally connected or commonly supported with sufficient rigidity such that binding of the operating shaft is avoided.
3. Any ceramic or other nonductile components in the seismic load path are specifically evaluated.
4. Adjacent electrical cabinets are bolted together and cabinet lineups are prevented from impacting adjacent structural members.

Components that could be damaged, or could damage other components, and are fastened to multiple locations of a structure, must be designed to accommodate seismic relative displacements. Such components include bus ducts, cable trays, conduits, elevator guide rails, and piping systems. As discussed in Section C13.3.2.1, special design consideration is required where full story drift demands are concentrated in a fraction of the story height.

The values of a_p and R_p for air coolers (commonly known as fin fans) with integral support legs in Table 13.6-1 are taken from *Guidelines for Seismic Evaluation and Design of Petrochemical Facilities* (ASCE 2011). The values listed for “fans” in Table 13.6-1 ($a_p = 2.5$ and $R_p = 6$) are not intended for fin fans with integral support legs. (They do apply where fin fans are not supported on integral support legs.) As discussed in ASCE (2011), fin fans with integral support legs have not performed



FIGURE C13.6-1 Fin Fan Elevated on Integral Supports

well in seismic events, such as the February 27, 2010, Chile earthquake.

Typically, fin fans are supported on pipe racks (Fig. C13.6-1). Where the fin fan is supported on legs, this configuration generally creates a condition where a relatively rigid mass is supported on flexible legs on top of a pipe rack and can result in significantly higher seismic force demands. The support legs should be braced in both directions. Knee braces should be avoided. Vertical bracing should intersect columns at panel points with beams. Where geometrically practical, chevron bracing may be used. Whenever possible, it is recommended that the fin fan should be designed without vendor-supplied integral legs and should be supported directly on the pipe rack structural steel. In such cases, the values of a_p and R_p for fans apply.

Regardless of whether the fin fan vendor or the engineering contractor provides the supporting steel, the structural steel directly supporting the air coolers should be designed to the same level of seismic detailing required of the pipe rack structural steel.

Mechanical components with similar construction details used in fin fans (such as air-cooled heat exchangers, condensing units, dry coolers, and remote radiators) are grouped with fin fans because similar behavior is assumed.

C13.6.4 Component Supports. The intent of this section is to require seismic design of all mechanical and electrical component supports to prevent sliding, falling, toppling, or other movement that could imperil life. Component supports are differentiated here from component attachments to emphasize that the supports themselves, as enumerated in the text, require seismic design even if they are fabricated by the mechanical or electrical component manufacturer. This need exists regardless of whether the mechanical or electrical component itself is designed for seismic loads.

C13.6.4.1 Design Basis. Standard supports are those developed in accordance with a reference document (Section 13.1.7). Where standard supports are not used, the seismic design forces and displacement demands of Chapter 13 are used with applicable material-specific design procedures of Chapter 14.

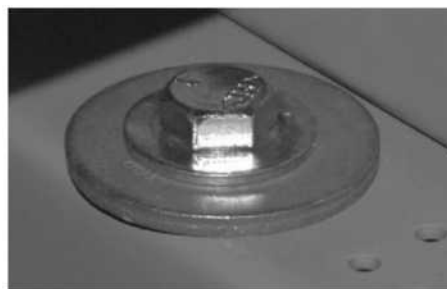
C13.6.4.2 Design for Relative Displacement. For some items, such as piping, seismic relative displacements between support points are of more significance than inertial forces. Components made of high-deformability materials such as steel or copper can accommodate relative displacements inelastically, provided that the connections also provide high deformability. Threaded and soldered connections exhibit poor ductility under inelastic displacements, even for ductile materials. Components made of less ductile materials can accommodate relative displacement effects only if appropriate flexibility or flexible connections are provided.

Detailing distribution systems that connect separate structures with bends and elbows makes them less prone to damage and less likely to fracture and fall, provided that the supports can accommodate the imposed loads.

C13.6.4.3 Support Attachment to Component. As used in this section, “integral” relates to the manufacturing process, not the location of installation. For example, both the legs of a cooling tower and the attachment of the legs to the body of the cooling tower must be designed, even if the legs are provided by the manufacturer and installed at the plant. Also, if the cooling tower has an $I_p = 1.5$, the design must address not only the attachments (e.g., welds and bolts) of the legs to the component but also local stresses imposed on the body of the cooling tower by the support attachments.



Failure of sheet metal base anchored with standard washer



Anchorage equipped with Belleville washer

FIGURE C13.6-2 Equipment Anchorage with Belleville Washers

Source: Courtesy of Philip Caldwell.

C13.6.4.5 Additional Requirements. As reflected in this section of the standard and in footnote b to Table 13.6-1, vibration-isolated equipment with snubbers is subject to amplified loads as a result of dynamic impact.

Most sheet metal connection points for seismic anchorage do not exhibit the same mechanical properties as bolted connections with structural elements. The use of Belleville washers improves the seismic performance of connections to equipment enclosures fabricated from sheet metal 7 gauge (0.18 in. (5 mm)) or thinner by distributing the stress over a larger surface area of the sheet metal connection interface, allowing for bolted connections to be torqued to recommended values for proper preload while reducing the tendency for local sheet metal tearing or bending failures or loosening of the bolted connection (Fig. C13.6-2). The intrinsic spring loading capacity of the Belleville washer assists with long-term preload retention to maintain integrity of the seismic anchorage.

Manufacturers test or design their equipment to handle seismic loads at the equipment “hard points” or anchor locations. The results of this design qualification effort are typically reflected in installation instructions provided by the manufacturer. It is imperative that the manufacturer’s installation instructions be followed. Where such guidance does not exist, the registered design professional should design appropriate reinforcement.

C13.6.5 Distribution Systems: Conduit, Cable Tray, and Raceways. The term *raceway* is defined in several standards with somewhat varying language. As used here, it is intended to describe all electrical distribution systems including conduit, cable trays, and open and closed raceways. Experience indicates that a size limit of 2.5 in. (64 mm) can be established for the provision of flexible connections to accommodate seismic relative displacements that might occur between pieces of connected equipment because smaller conduit normally possesses the required flexibility to accommodate such displacements. See additional commentary pertaining to exemption of trapeze-supported systems in Section C13.1.4.

C13.6.6 Distribution Systems: Duct Systems. Experience in past earthquakes has shown that HVACR duct systems are rugged and perform well in strong ground shaking. Bracing in accordance with ANSI/SMACNA 001 (2000) has been effective in limiting damage to duct systems. Typical failures have affected only system function, and major damage or collapse has been uncommon. Therefore, industry standard practices should prove adequate for most installations. Expected earthquake damage is limited to opening of duct joints and tears in ducts. Connection details that are prone to brittle failures, especially hanger rods subject to large amplitude cycles of

bending stress, should be avoided. See additional commentary in Section C13.1.4.

The amplification factor for duct systems has been increased from 1.0 to 2.5 because even braced duct systems are relatively flexible. The R_p values also have been increased so that the resulting seismic design forces are consistent with those determined previously.

Duct systems that carry hazardous materials or must remain operational during and after an earthquake are assigned a value of $I_p = 1.5$, and they require a detailed engineering analysis addressing leak tightness.

Lighter in-line components may be designed to resist the forces from Section 13.3 as part of the overall duct system design, whereby the duct attached to the in-line component is explicitly designed for the forces generated by the component. Where in-line components are more massive, the component must be supported and braced independently of the ductwork to avoid failure of the connections.

The requirements for flexible connections of unbraced piping to in-line components such as reheat coils applies regardless of the component weight.

C13.6.7 Distribution Systems: Piping and Tubing Systems. Because of the typical redundancy of piping system supports, documented cases of total collapse of piping systems in earthquakes are rare; however, pipe leakage resulting from excessive displacement or overstress often results in significant consequential damage and in some cases loss of facility operability. Loss of fluid containment (leakage) normally occurs at discontinuities such as threads, grooves, bolted connectors, geometric discontinuities, or locations where incipient cracks exist, such as at the toe or root of a weld or braze. Numerous building and industrial national standards and guidelines address a wide variety of piping systems materials and applications. Construction in accordance with the national standards referenced in these provisions is usually effective in limiting damage to piping systems and avoiding loss of fluid containment under earthquake conditions.

ASHRAE (2000) and MSS (2001) are derived in large part from the predecessors of SMACNA (2008). These documents may be appropriate references for use in the seismic design of piping systems. Because the SMACNA standard does not refer to pipe stresses in the determination of hanger and brace spacing, however, a supplementary check of pipe stresses may be necessary when this document is used. ASME piping rules as given in the ASME BPVC and ASME B31 parts B31.1, B31.3, B31.5, B31.9, and B31.12 are normally used for high-pressure, high-temperature piping but can also conservatively be applied to

other lower pressure, lower temperature piping systems. Code-compliant seismic design manuals prepared specifically for proprietary systems may also be appropriate references.

Table 13.6-1 entries for piping previously listed the amplification factor related to the response of piping systems as rigid ($a_p = 1.0$) and values for component response modification factors lower than in the current table. However, it was realized that most piping systems are flexible and that the amplification factor values should reflect this fact; thus, a_p was increased to 2.5 and the R_p values were adjusted accordingly such that a_p/R_p remains roughly consistent with earlier provisions.

Although seismic design in accordance with Section 13.6.8 generally ensures that effective seismic forces do not fail piping, seismic displacements may be underestimated such that impact with nearby structural, mechanical, or electrical components could occur. In marginal cases, it may be advisable to protect the pipe with wrapper plates where impacts could occur, including at gapped supports. Insulation may in some cases also serve to protect the pipe from impact damage. Piping systems are typically designed for pressure containment, and piping designed with a factor of safety of three or more against pressure failure (rupture) may be inherently robust enough to survive impact with nearby structures, equipment, and other piping, particularly if the piping is insulated. Piping that has less than standard weight wall thickness may require the evaluation of the effects of impact locally on the pipe wall and may necessitate means to protect the pipe wall.

It is usually preferable for piping to be detailed to accommodate seismic relative displacements between the first seismic support upstream or downstream from connections and other seismically supported components or headers. This accommodation is preferably achieved by means of pipe flexibility or, where pipe flexibility is not possible, flexible supports. Piping not otherwise detailed to accommodate such seismic relative displacements must be provided with connections that have sufficient flexibility in the connecting element or in the component or header to avoid failure of the piping. The option to use a flexible connecting element may be less desirable because of the need for greater maintenance efforts to ensure continued proper function of the flexible element.

Grooved couplings, ball joints, resilient gasket compression fittings, other articulating-type connections, bellows expansion joints, and flexible metal hose are used in many piping systems and can serve to increase the rotational and lateral deflection design capacity of the piping connections.

Grooved couplings are classified as either rigid or flexible. Flexible grooved couplings demonstrate limited free rotational capacity. The free rotational capacity is the maximum articulating angle where the connection behaves essentially as a pinned joint with limited or negligible stiffness. The remaining rotational capacity of the connection is associated with conventional joint behavior, and design force demands in the connection are determined by traditional means.

Rigid couplings are typically used for high-pressure applications and usually are assumed to be stiffer than the pipe. Alternatively, rigid coupling may exhibit bilinear rotational stiffness with the initial rotational stiffness affected by installation.

Coupling flexibilities vary significantly between manufacturers, particularly for rigid couplings. Manufacturer's data may be available. Industrywide procedures for the determination of coupling flexibility are not currently available; however, some guidance for couplings may be found in the provisions for fire sprinkler piping, where grooved couplings are classified as either rigid or flexible on the basis of specific requirements on angular movement. In Section 3.5.4 of NFPA (2007), flexible couplings are defined as follows:

A listed coupling or fitting that allows axial displacement, rotation, and at least 1 degree of angular movement of the pipe without inducing harm on the pipe. For pipe diameters of 8 in. (203.2 mm) and larger, the angular movement shall be permitted to be less than 1 degree but not less than 0.5 degrees.

Couplings determined to be flexible on this basis are listed either with FM Global (2007) or UL (2004).

Piping component testing suggests that the ductility capacity of carbon steel threaded and flexible grooved piping component joints ranges between 1.4 and 3.0, implying an effective stress intensification of approximately 2.5. These types of connections have been classified as having limited deformability, and piping systems with these connections have R_p values lower than piping with welded or brazed joints.

The allowable stresses for piping constructed with ductile materials assumed to be materials with high deformability, and not designed in accordance with an applicable standard or recognized design basis, are based on values consistent with industrial piping and structural steel standards for comparable piping materials.

The allowable stresses for piping constructed with low-deformability materials, and not designed in accordance with an applicable standard or recognized design basis, are derived from values consistent with ASME standards for comparable piping materials.

For typical piping materials, pipe stresses may not be the governing parameter in determining the hanger and other support spacing. Other considerations, such as the capacity of the hanger and other support connections to the structure, limits on the lateral displacements between hangers and other supports to avoid impacts, the need to limit pipe sag between hangers to avoid the pooling of condensing gases, and the loads on connected equipment, may govern the design. Nevertheless, seismic span tables, based on limiting stresses and displacements in the pipe, can be a useful adjunct for establishing seismic support locations.

Piping systems' service loads of pressure and temperature also need to be considered in conjunction with seismic inertia loads. The potential for low ambient and lower than ambient operating temperatures should be considered in the designation of the piping system materials as having high or low deformability. High deformability may often be assumed for steels, particularly ASME listed materials operating at high temperatures, copper and copper alloys, and aluminum. Low deformability should be assumed for any piping material that exhibits brittle behavior, such as glass, ceramics, and many plastics.

Piping should be designed to accommodate relative displacements between the first rigid piping support and connections to equipment or piping headers often assumed to be anchors. Barring such design, the equipment or header connection could be designed to have sufficient flexibility to avoid failure. The specification of such flexible connections should consider the necessity of connection maintenance.

Where appropriate, a walkdown of the finally installed piping system by an experienced design professional familiar with seismic design is recommended, particularly for piping greater than 6-in. (152.4-mm) nominal pipe size, high-pressure piping, piping operating at higher than ambient temperatures, and piping containing hazardous materials. The need for a walkdown may also be related to the scope, function, and complexity of the piping system, as well as the expected performance of the facility. In addition to providing a review of seismic restraint location, orientation, and attachment to the structure, the walkdown verifies that the required separation exists between the piping and nearby structures, equipment, and other piping in the as-built condition.

C13.6.7.1 ASME Pressure Piping Systems. In Table 13.6-1, the increased R_p values listed for ASME B31-compliant piping systems are intended to reflect the more rigorous design, construction, and quality control requirements, as well as the intensified stresses associated with ASME B31 designs.

Materials meeting ASME toughness requirements may be considered high-deformability materials.

C13.6.7.2 Fire Protection Sprinkler Piping Systems. The lateral design procedures of NFPA (2007) have been revised for consistency with the ASCE/SEI 7 design approach while retaining traditional sprinkler system design concepts. Using conservative upper bound values of the various design parameters, a single lateral force coefficient, C_p , was developed. It is a function of the mapped short-period response parameter S_s . Stresses in the pipe and connections are controlled by limiting the maximum reaction at bracing points as a function of pipe diameter.

Other components of fire protection systems, e.g., pumps and control panels, are subject to the general requirements of ASCE/SEI 7.

Experience has shown that interaction of other nonstructural components and sprinkler drops and sprigs is a significant source of damage and can result in serious consequential damage as well as compromise the performance of the fire protection system. Clearance for between sprinkler drops and sprigs and other nonstructural components needs to be addressed beyond NFPA 13. The minimum clearance value provided is based on judgment observations in past earthquakes. It is not the intent of this committee to require that sprinkler systems be field modified to accommodate these installed clearances if supports or equipment are installed after the sprinkler system is installed (i.e., the burden should not necessarily be on the sprinkler contractor to make the field modifications). It is the intent of this committee that the installation of permanently attached equipment, distribution systems, supports and fire sprinkler systems be coordinated such that the minimum clearance is maintained after their installation. As Building Information Systems become more widely used and nonstructural components and systems are detailed in the design phase of the project, maintaining these clearances should become easier to ensure by design.

C13.6.7.3 Exceptions. The conditions under which the force requirements of Section 13.3 may be waived are based on observed performance in past earthquakes. The limits on the maximum hanger or trapeze drop (hanger rod length) must be met by all the hangers or trapezes supporting the piping system. See additional commentary in Section C13.1.4.

C13.6.9 Utility and Service Lines. For essential facilities (Risk Category IV), auxiliary on-site mechanical and electrical utility sources are recommended.

Where utility lines pass through the interface of adjacent, independent structures, they must be detailed to accommodate differential displacement computed in accordance with Section 13.3.2 and including the C_d factor of Section 12.2.1.

As specified in Section 13.1.3, nonessential piping whose failure could damage essential utilities in the event of pipe rupture may be considered designated seismic systems.

C13.6.10 Boilers and Pressure Vessels. Experience in past earthquakes has shown that boilers and pressure vessels are rugged and perform well in strong ground motion. Construction in accordance with current requirements of the ASME *Boiler and Pressure Vessel Code* (ASME BPVC) has been shown to be effective in limiting damage to and avoiding loss of fluid containment in boilers and pressure vessels under earthquake conditions. It is, therefore, the intent of the standard

that nationally recognized codes be used to design boilers and pressure vessels provided that the seismic force and displacement demands are equal to or exceed those outlined in Section 13.3. Where nationally recognized codes do not yet incorporate force and displacement requirements comparable to the requirements of Section 13.3, it is nonetheless the intent to use the design acceptance criteria and construction practices of those codes.

C13.6.11 Elevator and Escalator Design Requirements. The ASME *Safety Code for Elevators and Escalators* (ASME A17.1) has adopted many requirements to improve the seismic response of elevators; however, they do not apply to some regions covered by this chapter. These changes are to extend force requirements for elevators to be consistent with the standard.

C13.6.11.3 Seismic Controls for Elevators. ASME A17.1 Section 8.4.10.1.2, specifies the requirements for the location and sensitivity of seismic switches to achieve the following goals: (a) safe shutdown in the event of an earthquake severe enough to impair elevator operations, (b) rapid and safe reactivation of elevators after an earthquake, and (c) avoidance of unintended elevator shutdowns. This level of safety is achieved by requiring the switches to be in or near the elevator equipment room, by using switches located on or near building columns that respond to vertical accelerations that would result from P and S waves, and by setting the sensitivity of the switches at a level that avoids false shutdowns because of nonseismic sources of vibration. The trigger levels for switches with horizontal sensitivity (for cases where the switch cannot be located near a column) are based on the experience with California hospitals in the Northridge earthquake of 1994. Elevators in which the seismic switch and counterweight derail device have triggered should not be put back into service without a complete inspection. However, in the case where the loss of use of the elevator creates a life-safety hazard, an attempt to put the elevator back into service may be attempted. Operating the elevator before inspection may cause severe damage to the elevator or its components.

The building owner should have detailed written procedures in place defining for the elevator operator and/or maintenance personnel which elevators in the facility are necessary from a post-earthquake, life-safety perspective. It is highly recommended that these procedures be in place, with appropriate personnel training, before an event occurs that is strong enough to trip the seismic switch.

C13.6.11.4 Retainer Plates. The use of retainer plates is a low-cost provision to improve the seismic response of elevators.

C13.6.12 Rooftop Solar Panels. Rooftop solar panels without positive attachment to the roof structure are limited to low-profile panels with a low height-to-depth ratio that respond by sliding on the roof surface without overturning. The amount of roof slope is limited because studies show that panels on sloped surfaces tend to displace in the downslope direction when subjected to seismic shaking, and the displacement increases with greater roof slope.

Displacement-based design of panels includes verifying that the panel remains safe if displaced. It needs to be verified that there is roof capacity to support the weight of the displaced panel and that wiring to the panel can accommodate the design panel displacement without damage.

Eq. (13.6-1) conservatively assumes a minimum coefficient of friction between the solar panel and the roof of 0.4. In cold-weather regions, the effects on the friction coefficient should be considered for Seismic Design Categories D, E, and F.

Structural interconnection between portions of a panel must be of adequate design strength, in tension or compression, and

stiffness in order to account for the potential that frictional resistance to sliding will be different under some portions of the panel as a result of varying normal force and actual instantaneous values of friction coefficient for a given roof surface material.

The requirement for unattached panel to be bounded by a curb or parapet is usually satisfied by a curb at the roof edge. In lieu of being bounded by curbs or parapets at roof edges and offsets, the panel may be set back a larger distance from the edge.

Analytical and experimental studies of the seismic response of unattached solar panels are reported by Schellenberg et al. (2012) and Maffei et al. (2013).

Shake table testing and nonlinear time history analysis may also be used to predict panel displacements; however, for unattached panels, it is necessary to use input motions appropriate for predicting sliding displacement, which can be affected by content in the low-frequency range. See SEAOC (2012) for guidance on the performance of such testing and analysis.

C13.6.13 Other Mechanical and Electrical Components. The material properties set forth in item 2 of this section are similar to those allowed in ASME BPVC and reflect the high factors of safety necessary for seismic, service, and environmental loads.

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